



RAJALAKSHMI
ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

DEPARTMENT OF INFORMATION TECHNOLOGY LAB MANUAL

CS23432 – Software Construction

(REGULATION 2023)

RAJALAKSHMI ENGINEERING COLLEGE
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INDEX

S.No	Date	Title
1.	21/01/26	Azure Devops Environment Setup.
2.	28/01/26	Azure Devops Project Setup and User Story Management.
3.	04/02/26	Setting Up Epics, Features, And User Stories for Project Planning.
4.	11/02/26	Sprint Planning.
5.	18/02/26	Poker Estimation.
6.	25/02/26	Designing Class and Sequence Diagrams for Project Architecture.
7.	11/03/26	Designing Architectural and ER Diagrams for Project Structure.
8.	18/03/26	Testing – Test Plans and Test Cases.
9.	25/03/26	Load Testing and Pipelines.
10.	08/04/26	GitHub: Project Structure & Naming Conventions.

EXP NO: 1	AZURE DEVOPS ENVIRONMENT SETUP
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Aim:

To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

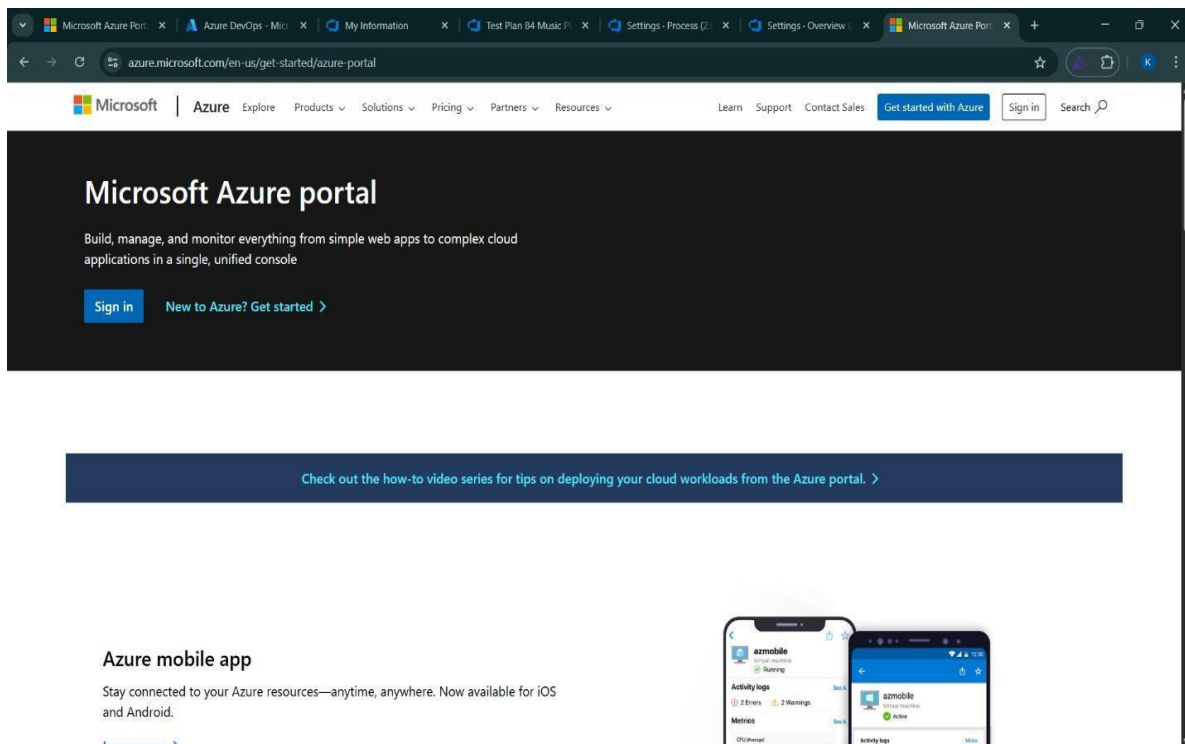
INSTALLATION:

1. Open your web browser and go to the Azure website:

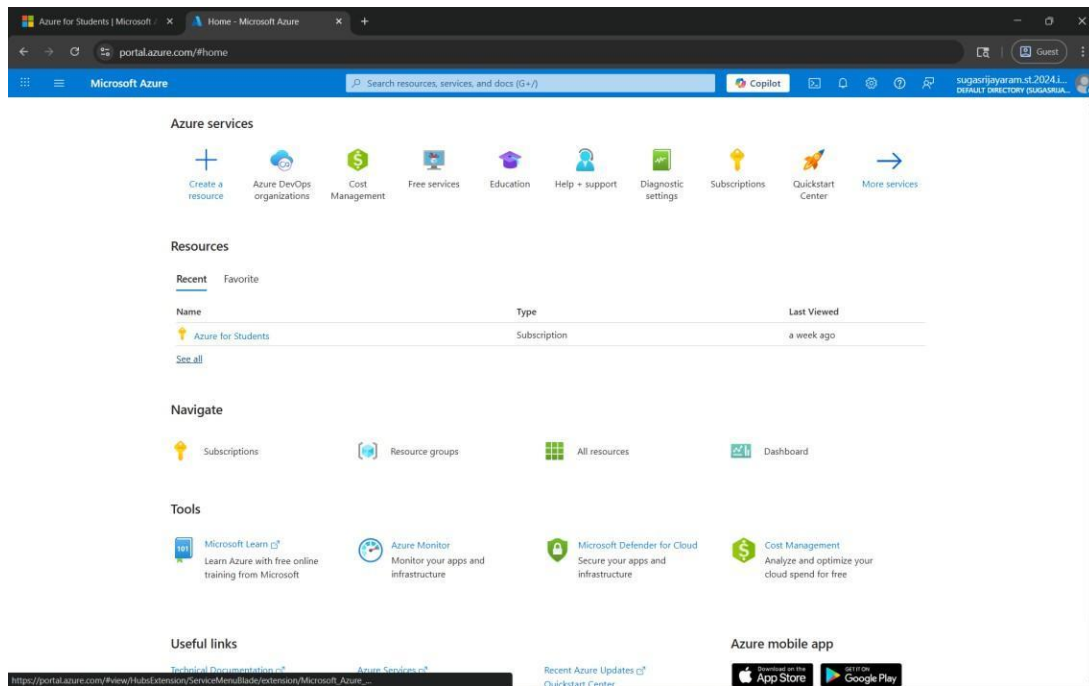
<https://azure.microsoft.com/en-us/get-started/azure-portal>.

Sign in using your Microsoft account credentials.

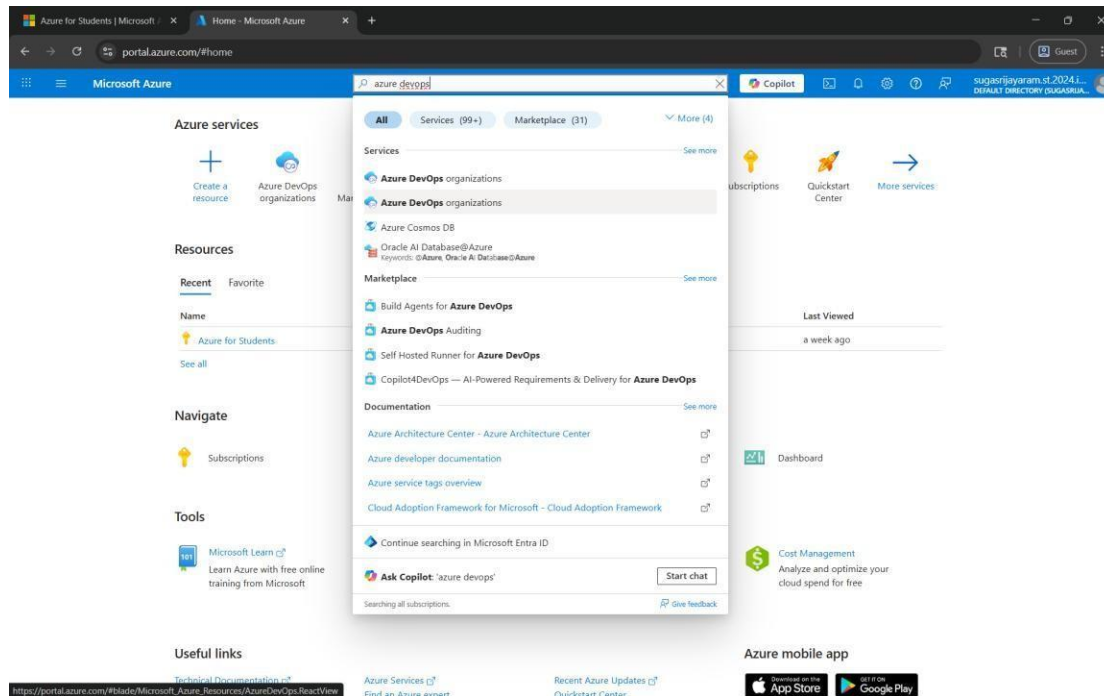
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>



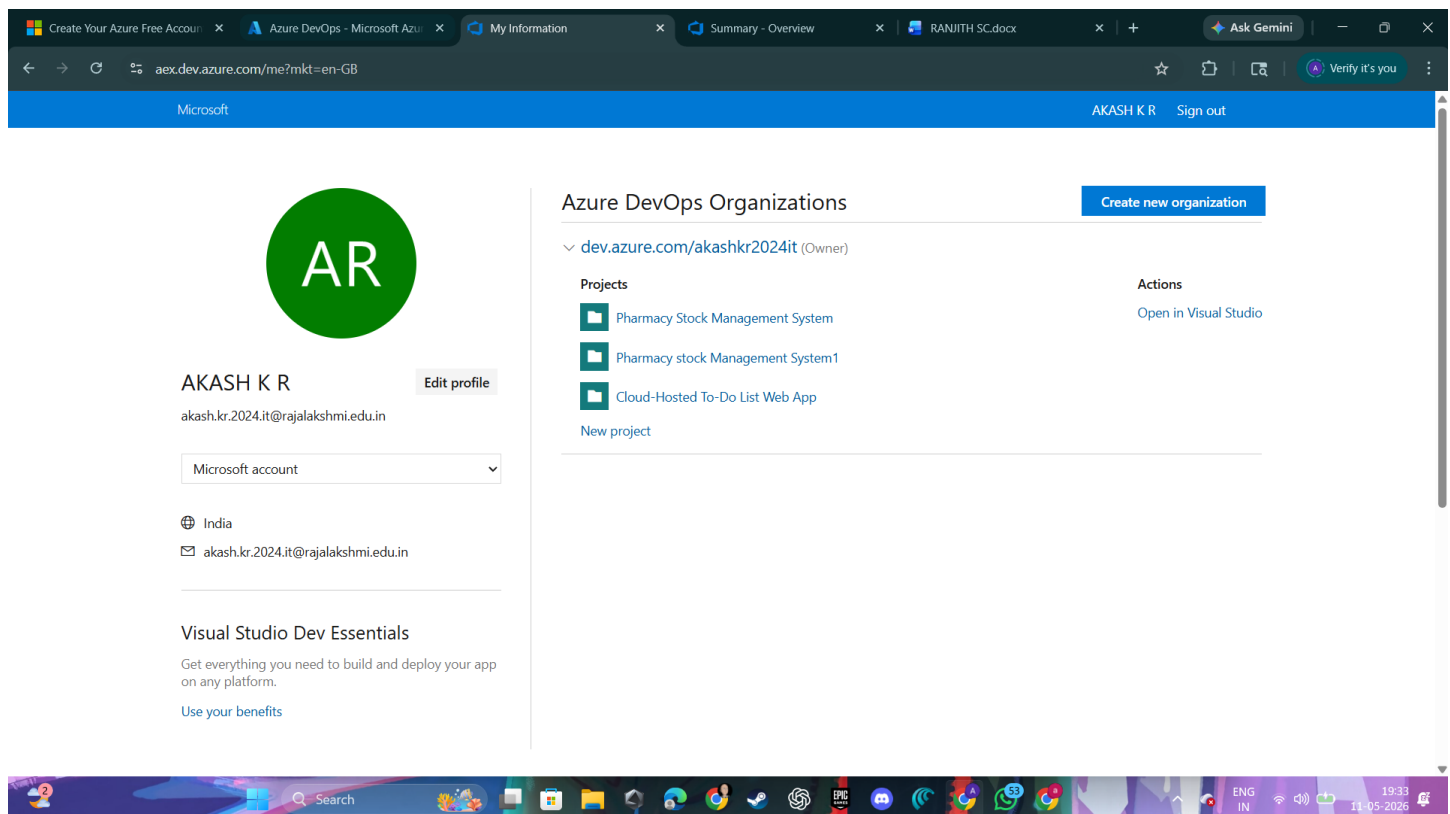
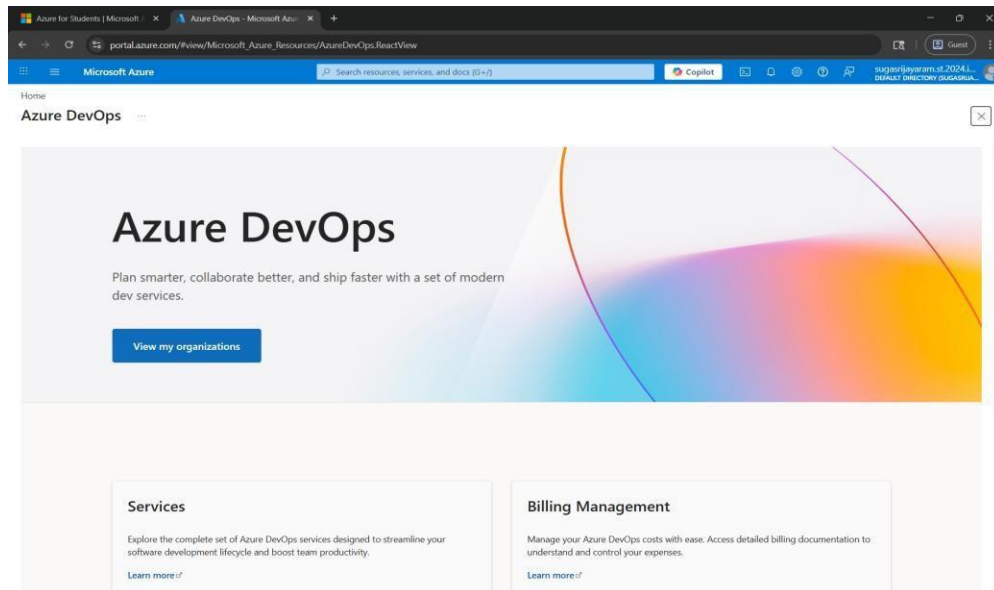
2. Azure home page



3. Open DevOps environment in the Azure platform by typing *Azure DevOps Organizations* in the search bar.



4. Click on the **View My Organization** link and create an organization and you should be taken to the Azure DevOps Organization Home page.



Result:

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

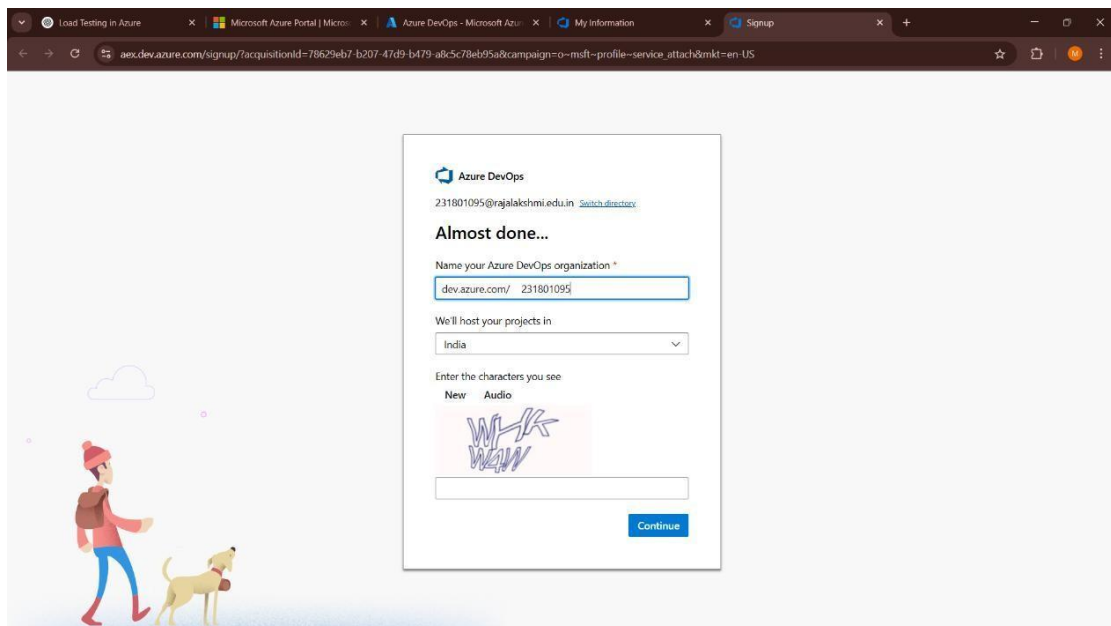
EXP NO: 2	AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT
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Aim:

To set up an Azure DevOps project for efficient collaboration and agile work management.

Procedure:

1.Create An Azure Account



2.Create the First Project in Your Organization

a. After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.

b. On the organization's **Home page**, click on the **New Project** button.

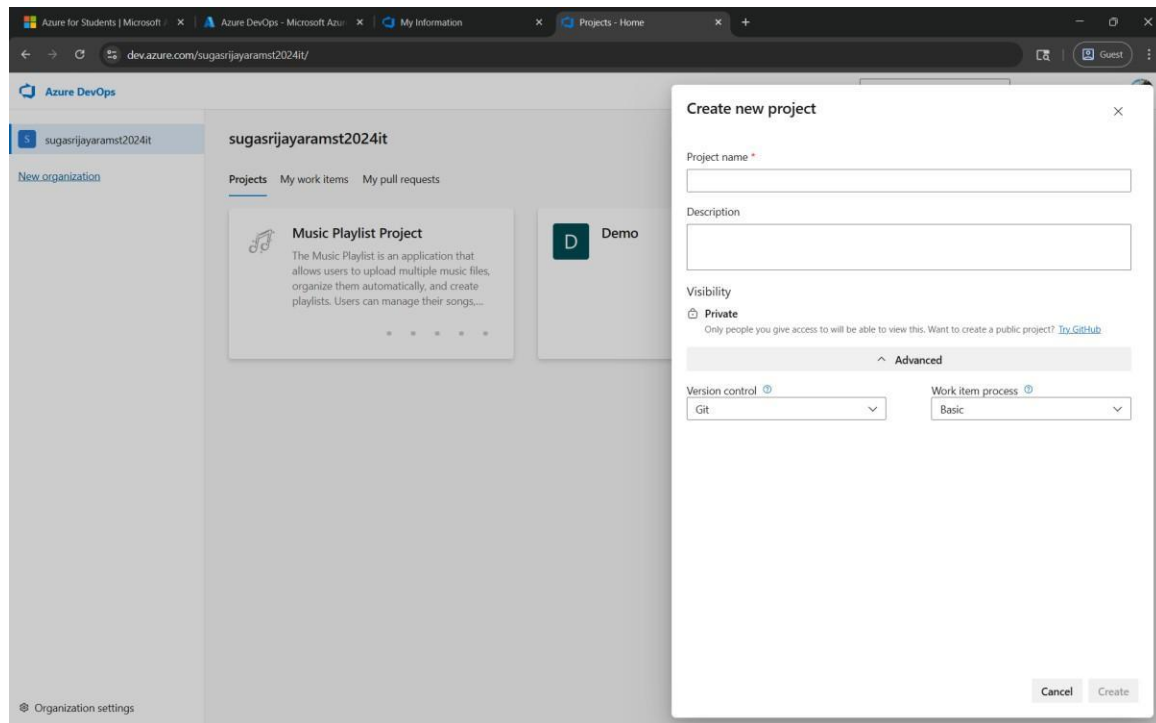
c. Enter the project name, description, and visibility options:

Name: Music Playlist Project.

Description: The Music Playlist is an application that allows users to upload multiple music files, organize them automatically, and create playlists. Users can manage their songs, arrange playlists, and download songs securely after payment. The system also uses external APIs to fetch song metadata and improve music organization.

Visibility: Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).

d. Once you've filled out the details, click **Create** to set up your first project.



3. Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.

Create Your Azure Free Account

Azure DevOps - Microsoft Azu

My Information

Summary - Overview

RANIITH SC.docx

Ask Gemini

Microsoft

AKASH K RSign out

AR

AKASH K R

Edit profile

akash.kr.2024.it@rajalakshmi.edu.in

Microsoft account

India

akash.kr.2024.it@rajalakshmi.edu.in

Visual Studio Dev Essentials

Get everything you need to build and deploy your app on any platform.

Use your benefits

Azure DevOps Organizations

Create new organization

dev.azure.com/akashkr2024it (Owner)

Projects

Pharmacy Stock Management System

Pharmacy stock Management System1

Cloud-Hosted To-Do List Web App

New project

Actions

Open in Visual Studio

4. Project dashboard

The screenshot shows the Azure DevOps interface for a project named 'Pharmacy stock Management System1'. The left-hand navigation menu includes Overview, Summary, Dashboards, Wiki, Boards, Repos, Pipelines, Test Plans, and Artifacts. The main content area displays 'About this project' with a detailed description of the system's objectives and features. The right-hand sidebar shows 'Project stats' for the last 7 days, 'Boards' with 16 work items, and 'Members' with 3 users.

About this project

The Pharmacy Management System is a modern web-based application designed to automate and simplify the various operations performed in a medical store or pharmacy. The main objective of the project is to reduce manual work, improve efficiency, maintain medicine records accurately, and provide a secure and user-friendly platform for pharmacy management. The system is developed using Python and Flask for backend development, while HTML, CSS, JavaScript, and Bootstrap are used to create an attractive and responsive user interface. The database connectivity is implemented using MongoDB or Microsoft Azure Cosmos DB, which stores all medicine details, billing records, sales information, and user credentials securely in the cloud. The project includes several important modules such as user authentication, medicine inventory management, billing system, dashboard analytics, stock monitoring, expiry tracking, and smart medicine search. The login system provides secure access for admins and normal users using role-based authentication, where admins are allowed to add, update, and delete medicine records while normal users have limited access permissions. The inventory management module helps maintain medicine stock details including medicine name, quantity, price, and expiry date, while automatically identifying low-stock medicines and expired products to improve stock control and prevent losses. The billing module generates bills automatically based on the selected medicines and quantities, calculates total prices, updates stock after purchase, and stores sales history for future reference. The dashboard module visually displays important statistics such as total medicines available, low-stock alerts, expired medicines, and total sales, helping the pharmacy management monitor business performance effectively. An AI-based smart medicine search feature is also included in the project, where users can search medicines based on symptoms such as fever, cold, pain, headache, and cough, making the system more intelligent and user-friendly. The project follows concepts of agile project management using epics, features, user stories, sprint planning, and poker planning implemented through Azure DevOps for project tracking and management. The application can be deployed locally or hosted on cloud platforms like Microsoft Azure App Service for online accessibility and scalability. Overall, the Pharmacy Management System is an efficient, secure, scalable, and easy-to-use solution that helps pharmacies manage daily operations digitally, reduce paperwork, improve billing accuracy, enhance inventory management, save time, and increase overall productivity while providing better service to customers.

5. To manage user stories:

- From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.
- On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.

The screenshot shows the 'Work items' page in the 'Boards' section of the Azure DevOps interface. The left-hand navigation menu includes Overview, Boards, Work items, Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, and Artifacts. The main content area displays a table of work items with columns for ID, Title, Assigned To, State, Area Path, and Tags. The table is filtered by keyword and shows 23 items.

ID	Title	Assigned To	State	Area Path	Tags
31	As a user, I want to search medicines using symptoms so that I can...	Unassigned	New	Pharmacy stock Management ...	
29	As a cashier, I want to generate bills automatically so that customer c...	Unassigned	New	Pharmacy stock Management ...	
26	As an admin, I want to log into the system securely so that I can mana...	yaswantkowsnik.c.2024.it...	New	Pharmacy stock Management ...	
30	As an admin, I want sales history reports so that business performanc...	Unassigned	New	Pharmacy stock Management ...	
28	As an admin, I want low-stock alerts so that medicines can be restock...	AKASH K R	New	Pharmacy stock Management ...	
27	As an admin, I want to add new medicines so that stock is updated in...	AKASH K R	New	Pharmacy stock Management ...	
18	Billing & Sales Management	ashok r	New	Pharmacy stock Management ...	
16	User Authentication	yaswantkowsnik.c.2024.it...	New	Pharmacy stock Management ...	
17	Inventory Management	AKASH K R	New	Pharmacy stock Management ...	
25	Smart Medicine Search	Unassigned	New	Pharmacy stock Management ...	
24	Sales Tracking	Unassigned	New	Pharmacy stock Management ...	
23	Generate Bills	Unassigned	New	Pharmacy stock Management ...	

USER STORY 3

3 Student Account Registration

No one selected

0 Comments Add Tag

Save and Close

Follow

State

New

Area

CEMS

Reason

Moved to the backlog

Iteration

CEMS

Updated by Ranjith Bharathwaj G. Mar 18

Details

1

0

Description

As a student, I want to create an account so that I can access the college event system.

Acceptance Criteria

- Registration form collects name, email, and password
- Email must be unique
- Data is stored successfully in the database
- Confirmation message is displayed

Discussion

86

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

switch to Markdown editor

Planning

Story Points

3

Priority

2

Risk

Classification

Value area

Business

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

Parent

2 User Registration

Updated Mar 17 New

Result:

Successfully created an Azure DevOps project with user story management and agile workflow setup.

EXP NO: 3

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

Aim:

To learn about how to create epics, user story, features, backlogs for college event management.

Create Epic, Features, User Stories, Task:

The screenshot displays the Azure DevOps interface for a project named 'Pharmacy stock Management'. The left sidebar shows the project structure with 'Work items' selected. The main area displays a list of work items under the 'Recently updated' filter. The work items are listed in a table with columns: ID, Title, Assigned To, State, Area Path, and Tags. The work items include user stories, features, and tasks related to a pharmacy stock management system.

ID	Title	Assigned To	State	Area Path	Tags
31	As a user, I want to search medicines using symptoms so that I can...	Unassigned	New	Pharmacy stock Management ...	
29	As a cashier, I want to generate bills automatically so that customer c...	Unassigned	New	Pharmacy stock Management ...	
26	As an admin, I want to log into the system securely so that I can mana...	yaswantkowschik.c.2024.it...	New	Pharmacy stock Management ...	
30	As an admin, I want sales history reports so that business performanc...	Unassigned	New	Pharmacy stock Management ...	
28	As an admin, I want low-stock alerts so that medicines can be restock...	AKASH K R	New	Pharmacy stock Management ...	
27	As an admin, I want to add new medicines so that stock is updated in	AKASH K R	New	Pharmacy stock Management ...	
18	Billing & Sales Management	ashok r	New	Pharmacy stock Management ...	
16	User Authentication	yaswantkowschik.c.2024.it...	New	Pharmacy stock Management ...	
17	Inventory Management	AKASH K R	New	Pharmacy stock Management ...	
25	Smart Medicine Search	Unassigned	New	Pharmacy stock Management ...	
24	Sales Tracking	Unassigned	New	Pharmacy stock Management ...	
23	Generate Bills	Unassigned	New	Pharmacy stock Management ...	

Fill in Epics

👑 EPIC 1

1 User Management

No one selected

0 Comments

Add Tag

State: New

Area: CEMS

Reason: New

Iteration: CEMS

Updated by Ranjith Bharathwaj G: Mar 17

Description

Click to add Description.

Discussion

NS

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

switch to Markdown editor

Planning

Priority: 2

Risk

Effort

Business Value

Time Criticality

Start Date: Select a date...

Target Date: Select a date...

Classification

Value area: Business

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

[Add an existing work item as a parent](#)

Child

👑 8 Profile Management
Updated Mar 17 • New

👑 5 User Authentication

Fill in Features

👑 FEATURE 2

2 User Registration

No one selected

0 Comments

Add Tag

State: New

Area: CEMS

Reason: New

Iteration: CEMS

Updated by Ranjith Bharathwaj G: Mar 17

Description

Click to add Description.

Discussion

NS

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

switch to Markdown editor

Planning

Priority: 2

Risk

Effort

Business Value

Time Criticality

Start Date: Select a date...

Target Date: Select a date...

Classification

Value area: Business

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

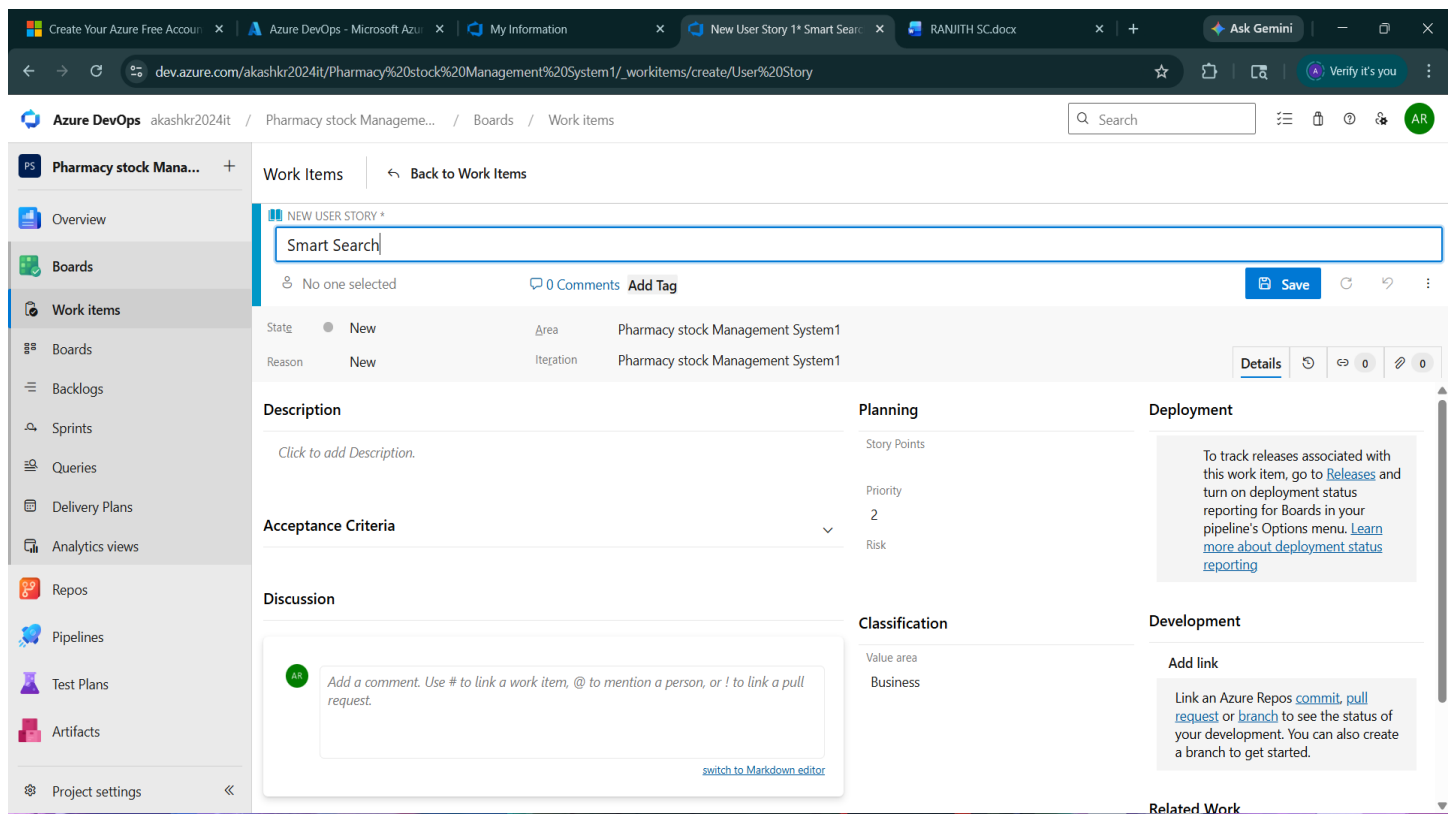
Parent

👑 1 User Management
Updated Mar 17 • New

Child

👑 4 Email Validation
Updated Mar 17 • New

Fill in User Story Detail



Result:

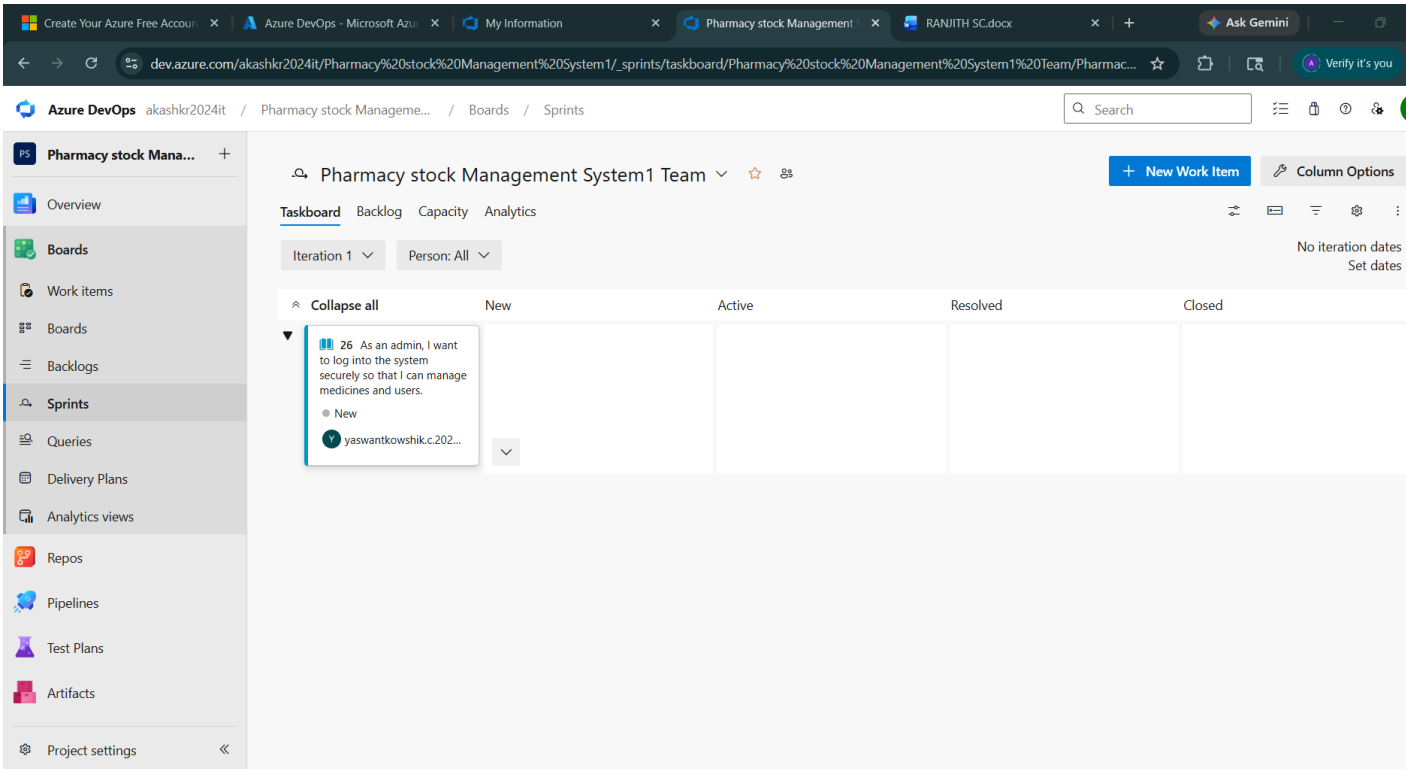
Thus, the creation of epics, features, user story and task has been created successfully.

EXP NO: 4	SPRINT PLANNING
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Aim:
To assign user story to specific sprint for the College event management Project.

Sprint Planning:

Sprint 1



Sprint 2

Create Your Azure Free Account

Azure DevOps - Microsoft Azure

My Information

Pharmacy stock Management

RANIITH SC.docx

Ask Gemini

Verify it's you

dev.azure.com/akashkr2024it/Pharmacy%20stock%20Management%20System1/_sprints/taskboard/Pharmacy%20stock%20Management%20System1%20Team/Pharmac...

Azure DevOps

akashkr2024it / Pharmacy stock Manage...

Boards / Sprints

Search

AR

PS Pharmacy stock Mana...

Overview

Boards

Work items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

Repos

Pipelines

Test Plans

Artifacts

Project settings

Pharmacy stock Management System1 Team

New Work Item

Column Options

Taskboard Backlog Capacity Analytics

Iteration 2 Person: All

No iteration dates Set dates

Collapse all	New	Active	Resolved	Closed
27 As an admin, I want to add new medicines so that stock is updated in the database. New AKASH K R				
28 As an admin, I want low-stock alerts so that medicines can be restocked before shortage occurs. New AKASH K R				

Sprint 3

Create Your Azure Free Account

Azure DevOps - Microsoft Azure

My Information

Pharmacy stock Management

RANJITH SC.docx

Ask Gemini

dev.azure.com/akashkr2024it/Pharmacy%20stock%20Management%20System1/_sprints/taskboard/Pharmacy%20stock%20Management%20System1%20Team/Pharmac...

Verify it's you

Azure DevOps

akashkr2024it / Pharmacy stock Manage...

Boards / Sprints

Search

AR

Pharmacy stock Mana...

Overview

Boards

Work items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

Repos

Pipelines

Test Plans

Artifacts

Project settings

Pharmacy stock Management System1 Team

New Work Item

Column Options

Taskboard Backlog Capacity Analytics

Iteration 3 Person: All

No iteration dates Set dates

Collapse all

New	Active	Resolved	Closed
29 As a cashier, I want to generate bills automatically so that customer checkout becomes faster. New Unassigned			
30 As an admin, I want sales history reports so that business performance can be analyzed. New Unassigned			
31 As a user, I want to search medicines using symptoms so that I can find suitable medicines quickly. New			

Sprint 4

Create Your Azure Free Account

Azure DevOps - Microsoft Azure

My Information

Pharmacy stock Management

RANJITH SC.docx

Ask Gemini

dev.azure.com/akashkr2024it/Pharmacy%20stock%20Management%20System1/_sprints/taskboard/Pharmacy%20stock%20Management%20System1%20Team/Pharmac...

Verify it's you

Azure DevOps

akashkr2024it / Pharmacy stock Manage...

Boards / Sprints

Search

AR

Pharmacy stock Mana...

Overview

Boards

Work items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

Repos

Pipelines

Test Plans

Artifacts

Project settings

Pharmacy stock Management System1 Team

New Work Item

Column Options

Taskboard Backlog Capacity Analytics

Iteration 3 Person: All

No iteration dates Set dates

Collapse all

New	Active	Resolved	Closed
29 As a cashier, I want to generate bills automatically so that customer checkout becomes faster. New Unassigned			
30 As an admin, I want sales history reports so that business performance can be analyzed. New Unassigned			
31 As a user, I want to search medicines using symptoms so that I can find suitable medicines quickly. New			

A screenshot of the Azure DevOps web interface showing a Sprints taskboard for the 'Pharmacy stock Management System1 Team'.

The browser tabs at the top include: 'Create Your Azure Free Account', 'Azure DevOps - Microsoft Azure', 'My Information', 'Pharmacy stock Management', 'RANJITH SC.docx', and 'Ask Gemini'.

The URL bar shows: `dev.azure.com/akashkr2024it/Pharmacy%20stock%20Management%20System1/_sprints/taskboard/Pharmacy%20stock%20Management%20System1%20Team/Pharmac...`

The Azure DevOps header shows the user 'akashkr2024it' and the path 'Pharmacy stock Manageme... / Boards / Sprints'. A search bar is on the right.

The left sidebar contains the following navigation items: 'Pharmacy stock Mana...', 'Overview', 'Boards', 'Work items', 'Boards', 'Backlogs', 'Sprints' (selected), 'Queries', 'Delivery Plans', 'Analytics views', 'Repos', 'Pipelines', 'Test Plans', 'Artifacts', and 'Project settings'.

The main area is titled 'Pharmacy stock Management System1 Team'. It has tabs for 'Taskboard', 'Backlog', 'Capacity', and 'Analytics'. The 'Taskboard' tab is active.

Below the tabs, there are filters for 'Iteration 2' and 'Person: All'. A 'New Work Item' button and a 'Column Options' link are in the top right.

The taskboard is a Kanban-style board with columns: 'New', 'Active', 'Resolved', and 'Closed'. There are two work items in the 'New' column:

- Item 27: 'As an admin, I want to add new medicines so that stock is updated in the database.' (New)
- Item 28: 'As an admin, I want low-stock alerts so that medicines can be restocked before shortage occurs.' (New)

Both items are assigned to 'AKASH K R'.

Result:

The Sprints are created for the College event management.

EXP NO: 5	POKER ESTIMATION
------------------	-------------------------

Aim:

Create Poker Estimation for the user stories for Music Playlist Project.

Planning Poker Estimation:

User Story ID: 20

Title: User Registration and Authentication

Description:

As a user, I want to register and log in securely so that I can access my personal music playlists and downloads.

Story Point Estimation: 8 Points

Team Poker Votes:

Team Member	Vote (Story Points)
Team Member 1	5
Team Member 2	8
Team Member 3	8
Team Member 4	8

Final Consensus: 8 Points

Reason for Estimation:

- Includes multiple features: registration, login, session, logout
- Requires frontend + backend + database integration
- Security implementation (password hashing, authentication)
- Moderate complexity with high risk

Risks:

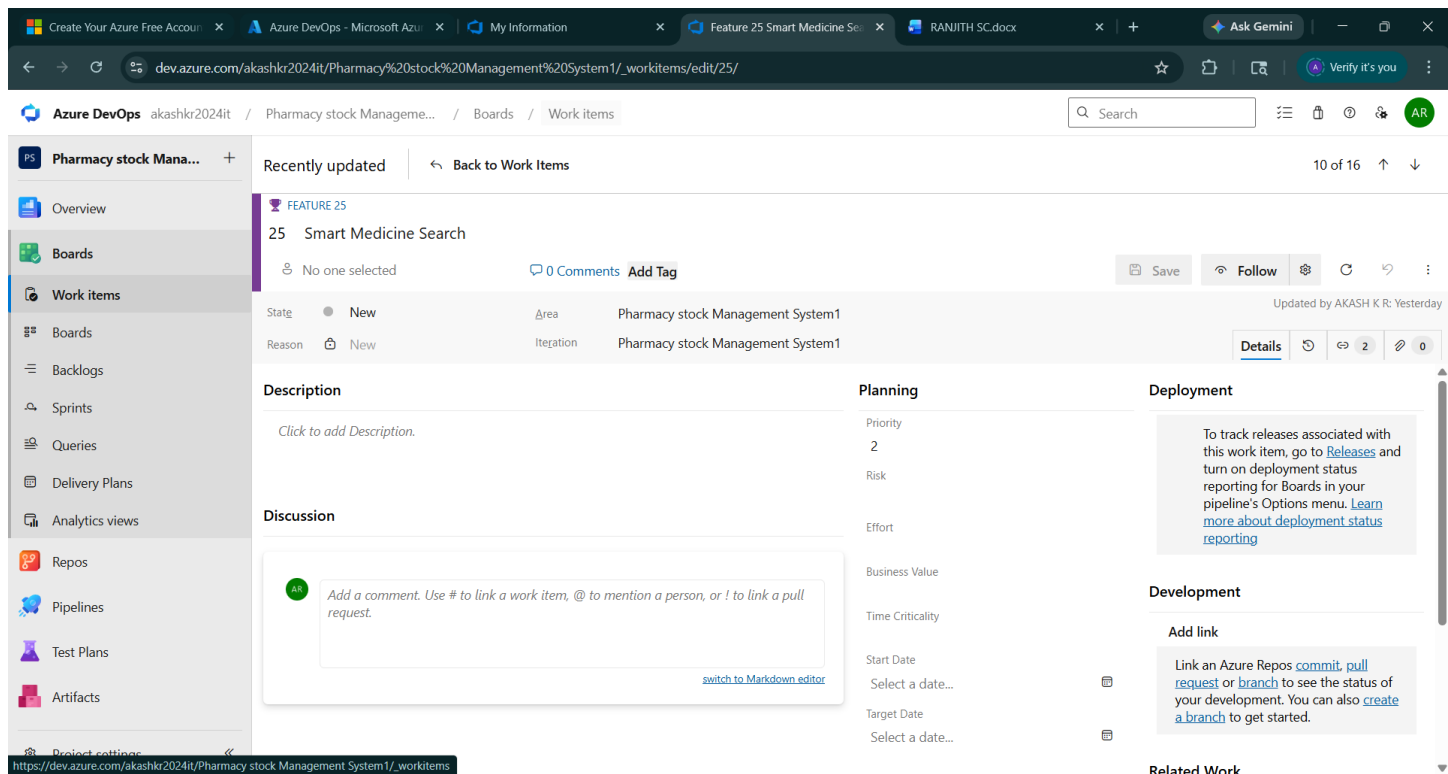
- Security vulnerabilities
- Session management issues
- Integration challenges

Acceptance Criteria Covered:

- User registration with email & password
- Login with valid credentials
- Error message for invalid login
- Session maintained after login
- Logout functionality

Conclusion:

Final agreed estimation is **8 story points** based on complexity, effort, and risk.



Result:

The Estimation/Story Points is created for the project using Poker Estimation.

EXP NO: 6	DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE
------------------	---

Aim:

To Design a Class Diagram and Sequence Diagram for the College Event Management.

Class Diagram:

Classes:

- Student
- Admin
- Event
- Registration
- Payment
- Certificate
- Notification
- Coordinator

Key Attributes:

Student:

StudentId,Name,Email,Department

Event:

EventId,Title,Date,Venue,Category,Fee

Registration:

RegistrationId,StudentId,E

ventId,PaymentStatus

Payment:

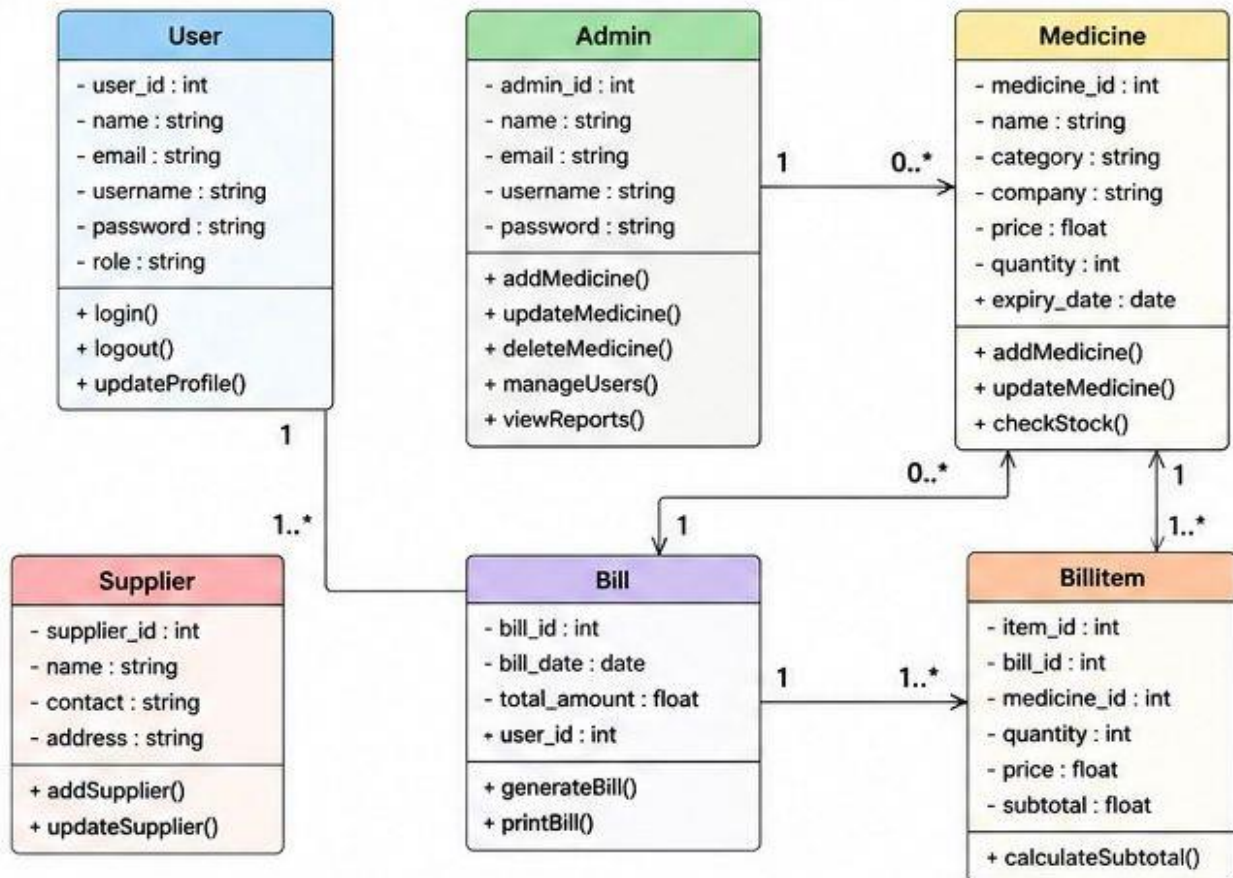
PaymentId,Amount,Tran

sactionStatus

Relationships:

- Student → registers for → Event
- Activity → conducted at → Venue
- Student → participates in → Competition
- Faculty Coordinator → manages → Event
- Organizer → schedules → Event
- Event → has → Participants
- Student → makes → Payment
- Student → receives → Ticket/Pass
- Student → submits → Feedback
- Student → receives → Notification
- Sponsor → sponsors → Event
- Event → awards → Certificate/Prize
- Department → organizes → Event
- Venue → allocated to → Event
- Student → joins → Team
- Team → participates in → Event
- Admin → approves → Event Request
- Event → publishes → Results
- Offer/Discount → applies to → Event Registration

CLASS DIAGRAM – PHARMACY MANAGEMENT SYSTEM



Sequence Diagram:

Actors:

User → Frontend → Backend → Database → Auth Service

Steps:

1. User enters email and password
2. Frontend validates input (email format, password)
3. Frontend sends login request to backend
4. Backend validates request data
5. Backend checks user in database

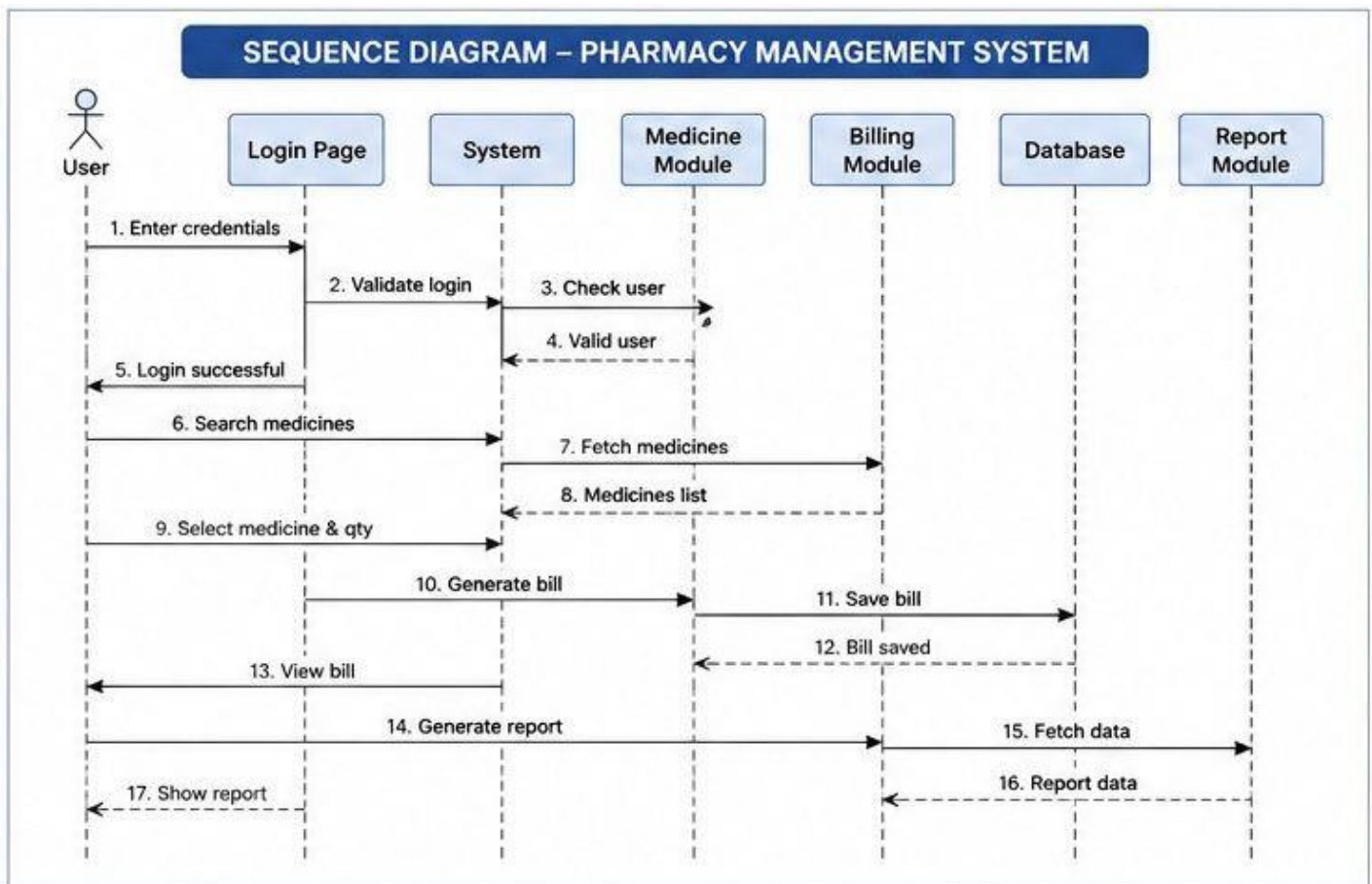
Condition: User Not Found

- Backend returns error: *Invalid credentials*

Condition: User Found

6. Backend verifies password
7. If password incorrect → return error
8. If password correct → proceed
9. Backend verifies user with auth service
10. Backend generates JWT token
11. Backend sets session (token expiry)
12. Backend sends success response

13. Frontend stores token
14. Frontend redirects to dashboard
15. User login successful



Result:The Class Diagram and Sequence Diagram is designed Successfully for the College Event Managemnt.

EXP NO: 7	DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE
------------------	--

Aim:

To Design an Architectural Diagram and ER Diagram for the given Project.

Architectural Diagram:

Layers:

1. Client Layer

- Browser
- React App
- Zustand (State Management)

2. API Communication

- Axios (HTTP Requests)

3. Backend Layer

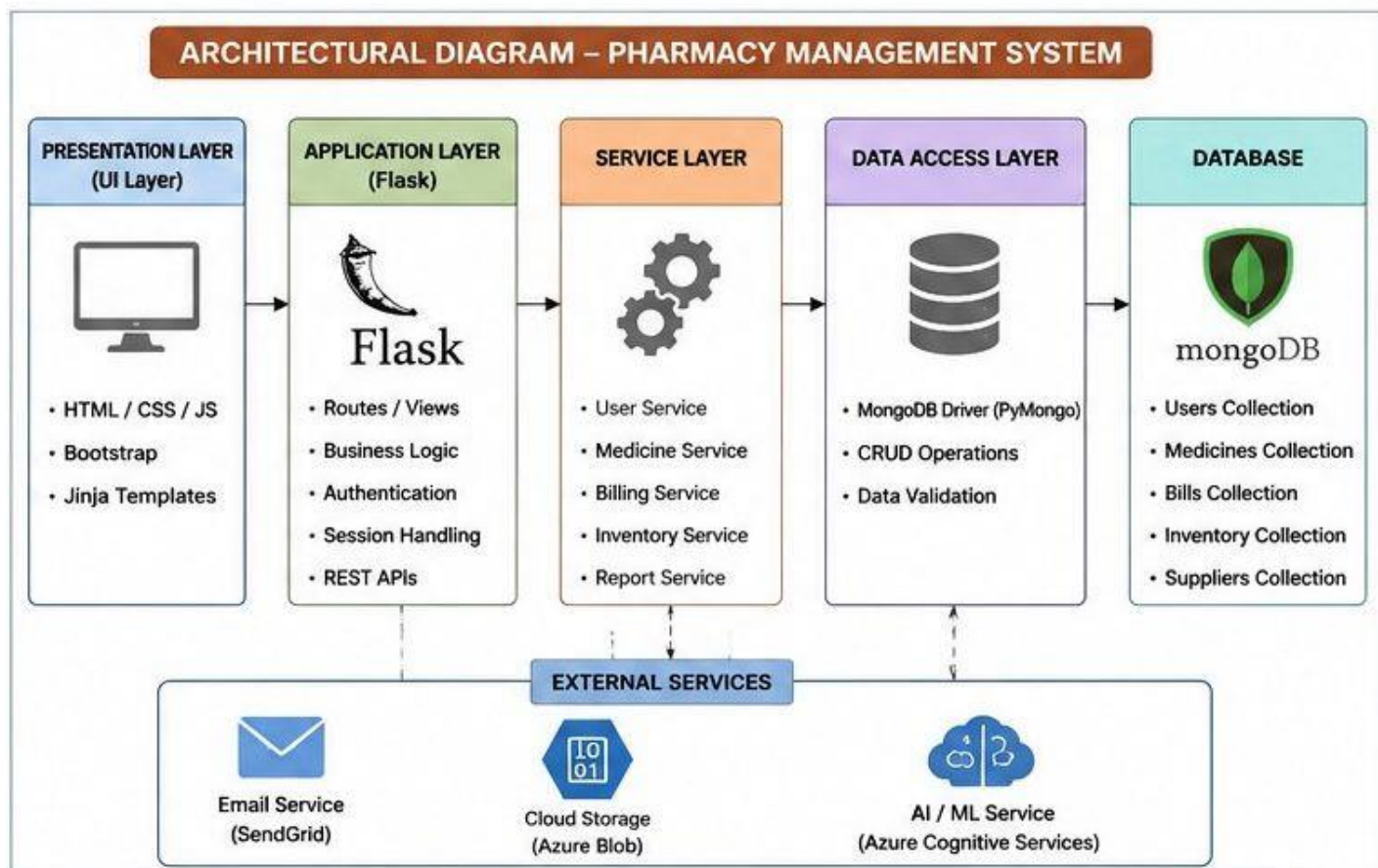
- Express Server
- Routes
- Controllers

4. Services

- Clerk Authentication
- File Upload (Cloudinary)

5. Database Layer

- Mongoose Models
- MongoDB



ER Diagram:

Entities & Attributes:

Student

- StudentId
- Name
- Email
- Department

Event:

- EventId
- Title
- Venue

- Date
 - Fee
-

Registration

- registration
 - amount
 - status
-

Payment

- paymentId
 - amount
 - status
-

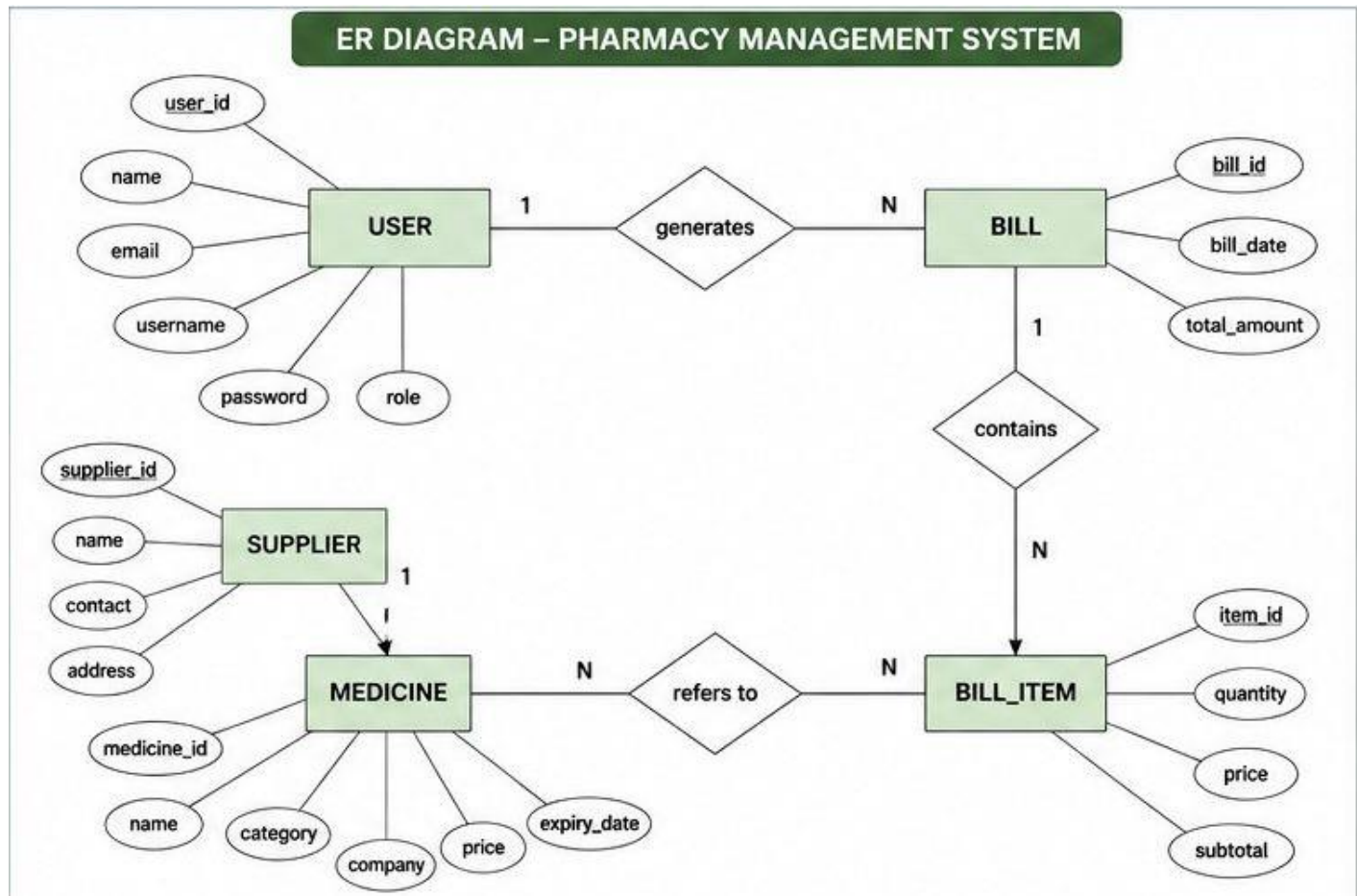
certificate

- certificateId

- issueDate
-

Relationships

- Student registers for Event (M : N)
- Event includes Activity (1 : M)
- Activity conducted at Venue (M : 1)
- Student participates in Competition (M : N)
- Faculty Coordinator manages Event (1 : M)
- Organizer schedules Event (1 : M)
- Event has Participants (1 : M)
- Student makes Payment (1 : M)
- Student receives Ticket / Pass (1 : 1)
- Student submits Feedback (1 : M)
- Student receives Notification (1 : M)
- Sponsor sponsors Event (M : N)
- Event awards Certificate / Prize (1 : M)
- Department organizes Event (1 : M)
- Venue allocated to Event (1 : M)
- Student joins Team (M : N)
- Team participates in Event (M : N)
- Admin approves Event Request (1 : M)
- Event publishes Results (1 : 1)
- Offer / Discount applies to Event Registration (1 : M)



Result: The Architecture Diagram and ER Diagram is designed Successfully for the College Event Management.

EXP NO: 8	TESTING – TEST PLANS AND TEST CASES
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Aim:

Test Plans and Test Case and write two test cases for at least two user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case:

Test Case Design Procedure

1. Understand Core Features of the Application

- User Signup & Login
- Viewing and Managing Playlists
- Fetching Real-time Metadata
- Editing playlists (rename, reorder, record)
- Creating smart audio playlists based on categories (mood, genre, artist, etc.)

2. Define User Interactions

- Each test case simulates a real user behaviour (e.g., logging in, renaming a playlist, adding a song).

3. Design Happy Path Test Cases

- Focused on validating that all features function as expected under normal conditions.
- Example: User logs in successfully, adds item to playlist, or creates a category-based playlist.

4. Design Error Path Test Cases

- Simulate negative or unexpected scenarios to test robustness and error handling.
- Example: Login fails with invalid credentials, save fails when offline, no recommendations found.

5. Break Down Steps and Expected Results

- Each test case contains step-by-step actions and a corresponding expected outcome.
- Ensures clarity for both testers and automation scripts.

6. Use Clear Naming and IDs

- Test cases are named clearly (e.g., TC01 – Successful signup, TC06 – Add and Remove Songs from Playlist).
- Helps in quick identification and linking to user stories or features.

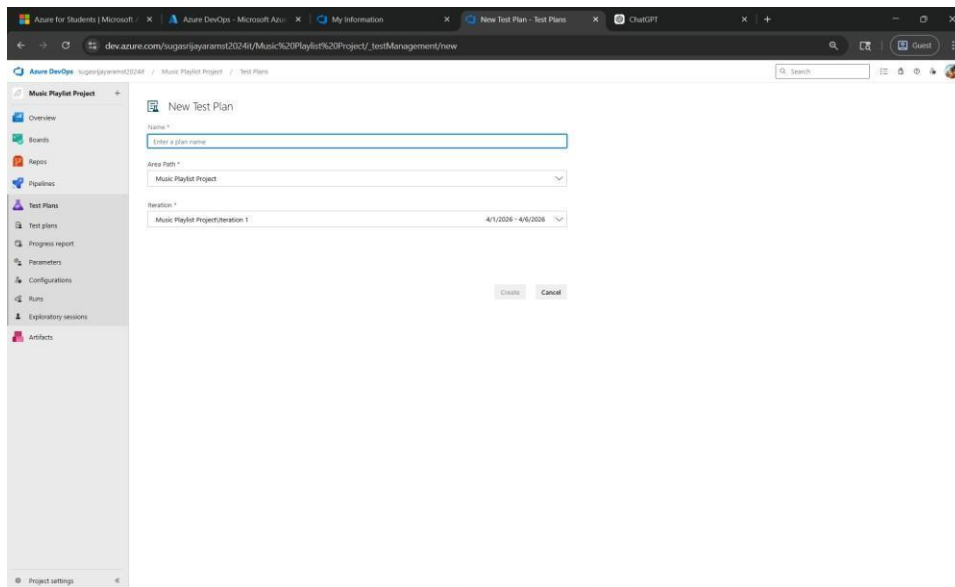
7. Separate Test Suites

- Grouped test cases based on functionality (e.g., Login, Playlist Editing, Recommendation System).
- Improves organization and test execution flow in Azure DevOps.

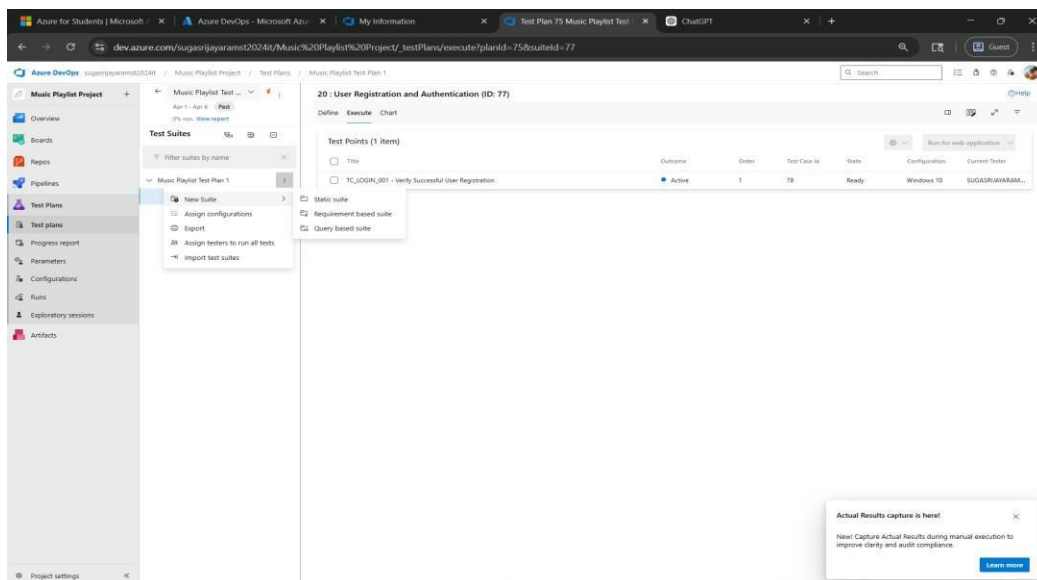
8. Prioritize and Review

- Critical user actions are marked high-priority.
- Reviewed for completeness and traceability against feature requirements.

1.New test plan



2.Test suite



3. Test case

Give two test cases for at least two user stories showcasing the happy path and error scenarios in azure DevOps platform.

Music Playlist Project – Test Plans

USER STORIES

- As a user, I want to register and log in securely so that I can access my personal music playlists and downloads. (ID: 20).
- As a user, I want to create and manage playlists so that I can organize my favourite songs easily (ID: 23).

Test Suites

Test Suit: TS01 - User Registration and Authentication (ID: 20)

1. TC01 – User Registration Successfully w/o Error

- **Action:**
 - Go to the Sign-Up page
 - Enter valid name, email, and password
 - Click "Sign Up"
- **Expected Results:**
 - Sign-Up form is displayed
 - Fields accept values without error
 - Account is created successfully
 - User is redirected to dashboard
- **Type: Happy Path**

2. TC02 – User Registration Failed with Invalid User Details

- **Action:**
 - Go to the Sign-Up page
 - Enter invalid details (empty fields / wrong email format / weak password)
 - Click "Sign Up"
- **Expected Results:**
 - Validation errors are displayed
 - Account is not created
- **Type: Error Path**

3. TC03 – User Login Successfully with Valid Credentials

- **Action:**
 - Go to Login page
 - Enter valid email and password
 - Click "Login"
- **Expected Results:**
 - Login form is displayed
 - Credentials are accepted
 - User is redirected to dashboard
- **Type: Happy Path**

4. TC04 – Login with Invalid Credentials

- **Action:**
 - Go to Login page
 - Enter invalid email or password
 - Click "Login"
- **Expected Results:**
 - Input is accepted
 - Error message "Invalid username or password" is shown
- **Type: Error Path**

Test Suit: TS02 - Playlist Creation and Management (ID: 23)

1. TC05 – Create Playlist Successfully

- **Action:**
 - Login with valid credentials
 - Navigate to "My Playlists"
 - Click "Create Playlist"
 - Enter playlist name
 - Click "Save/Create"
- **Expected Results:**
 - Playlist is created successfully
 - Playlist appears in list
- **Type: Happy Path**

2. TC06 – Add and Remove Songs from Playlist

- **Action:**
 - Open an existing playlist
 - Click "Add Song" and select a song
 - Click "Remove" on a song
- **Expected Results:**
 - Song is added to playlist
 - Song is removed from playlist
- **Type:** Happy Path

3. TC08 – Delete Playlist and Save Changes

- **Action:**
 - Navigate to "My Playlists"
 - Click "Delete" on a playlist
 - Confirm deletion
- **Expected Results:**
 - Playlist is deleted successfully
 - Playlist is removed from list
 - Changes are saved

Type: Happy Path

Test Cases

The screenshot displays the Azure DevOps Test Plans interface. The left sidebar shows the project structure with 'Test Plans' selected. The main area shows a test plan titled '20 : User Registration and Authentication (ID: 77)'. The test plan is in a 'Pass' state, indicating 100% success. Below the test plan details, there is a table of test cases.

Title	Order	Test Case Id	Assigned To	State
TC01 – User Registration Successfully w/o Error	1	78	SUGASRUJAYAR...	Closed
TC02 – User Registration Failed with Invalid User Details	2	79	SUGASRUJAYAR...	Closed
TC03 – User Login Successfully with Valid Credentials	3	80	Sridhar C.	Closed
TC04-Login with invalid credentials	4	81	Sridhar C.	Closed

TEST CASE 78

78 TC01 – User Registration Successfully w/o Error

SUGASRIJAYARAM S T

0 Comments Add Tag

State: Closed Area: Music Playlist Project Reason: Obsolete Iteration: Music Playlist Project\Iteration 1

Updated by SUGASRIJAYARAM S T: 1h ago

Steps Summary Associated Automation

Steps

Steps	Action	Expected result	Attachments
1.	Navigate to web page	Login page loads	
2.	Click "Get Started"	Registration form appears	
3.	Enter First name and last name	Name fields accept input	
4.	Enter Username	Username field accepts input	
5.	Enter valid email address	Email field accepts input	
6.	Enter password	Password field accepts input	
7.	Click "Continue" button	Account is created	
8.	Check email and enter verification code	Verification is successful	
9.	Click "Continue"	User is redirected to Home page	

Click or type here to add a step

Parameter values

Recent test results

- Passed by SUGASRIJAYARAM S T with Windows 10 Completed 12:07 PM, 4/28/2026
- Failed by SUGASRIJAYARAM S T with Windows 10 Completed 12:06 PM, 4/28/2026
- Passed by SUGASRIJAYARAM S T with Windows 10 Completed 3:20 AM, 4/28/2026

[View all test result history](#)

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

Azure DevOps

sugasrijayaramst2024it / Music Playlist Project / Test Plans / Music Playlist Test Plan 1

Overview Boards Repos Pipelines Test Plans Test plans Progress report Parameters Configurations Runs Exploratory sessions Artifacts Project settings

Music Playlist Test Plan 1

Apr 1 - Apr 6 Past 100% run, 100% passed. [View report](#)

Test Suites

Filter suites by name

Music Playlist Test Plan 1

- 23 : Playlist Creation and Ma...
- 20 : User Registration and A...

20 : User Registration and Authentication

Define Execute Chart

Test Points (4 items)

Title
☒ TC01 – User Registration Successfully w/o Error
☐ TC02 – User Registration Failed with Invalid Email
☐ TC03 – User Login Successfully with Valid Credentials
☐ TC04 -Login with invalid credentials

Test point ID 2 execution history

Test Case: 78 TC01 – User Registration Successfully w/o Error

Test Suite: 77 20 : User Registration and Authentication

Current Tester: SUGASRIJAYARAM S T

Configuration: Windows 10

Outcome	Test run	Run by	Time completed
Passed	9	SUGASRIJAYARAM S...	Yesterday
Failed	8	SUGASRIJAYARAM S...	Yesterday
Passed	5	sriram S	Yesterday

Currently showing a test point execution history. View all history for a full test case execution history. [View all history](#)

TEST CASE 81

81 TC04-Login with invalid credentials

Sridhar C 0 Comments Add Tag

Save and Close Follow

Updated by SUGASRIJAYARAM S T: 42m ago

State: Closed Area: Music Playlist Project Reason: Obsolete Iteration: Music Playlist Project\Iteration 1

Steps Summary Associated Automation 2 0

Steps

Steps	Action	Expected result	Attachments
1.	Open login page	Login page is displayed	
2.	Enter invalid username/email	Input is accepted	
3.	Enter invalid password	Input is accepted	
4.	Click "Login"	System validates credentials	
5.	Login attempt fails	Error message displayed: "Invalid credentials"	

Click or type here to add a step

Recent test results

Passed by Sridhar C with Windows 10 Completed 12:28 PM, 4/28/2026

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

Parent

20 User Registration and Authentication Updated Yesterday Resolved

Tests

20 User Registration and Authentication

TEST CASE 85

85 TC06 - Add and Remove Songs from Playlist

SUGASRIJAYARAM S T 0 Comments Add Tag

Save and Close Follow

Updated by SUGASRIJAYARAM S T: 43m ago

State: Closed Area: Music Playlist Project Reason: Obsolete Iteration: Music Playlist Project\Iteration 1

Steps Summary Associated Automation 2 0

Steps

Steps	Action	Expected result	Attachments
1.	Open an existing playlist	Playlist details are displayed	
2.	Click "Add Song"	Song selection options appear	
3.	Select a song	Song is added to the playlist	
4.	Verify playlist	Added song is visible in the list	
5.	Click "Remove/Delete" on a song	Song is removed	
6.	Verify playlist	Removed song no longer appears	

Click or type here to add a step

Recent test results

Passed by SUGASRIJAYARAM S T with Windows 10 Completed 4:36 PM, 4/29/2026

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

Parent

23 Playlist Creation and Management Updated 54m ago Resolved

Tests

23 Playlist Creation and Management

4. Running the test cases

23 : Playlist Creation and Management (ID: 83)

Define Execute Chart

Test Points (3 items)

Title	Outcome	Order	Test Case Id	Status
TC05 - Create Playlist Successfully	Passed	1	84	Completed
TC06 - Add and Remove Songs from Playlist	Passed	2	85	Completed
TC08 - Delete Playlist and Save Changes	Passed	3	86	Completed

Context menu options:

- View execution history
- Mark Outcome
- Run
- Reset test to active
- Edit test case
- Assign tester
- View test result

Run options:

- Run for web application
- Run for desktop application
- Run with options

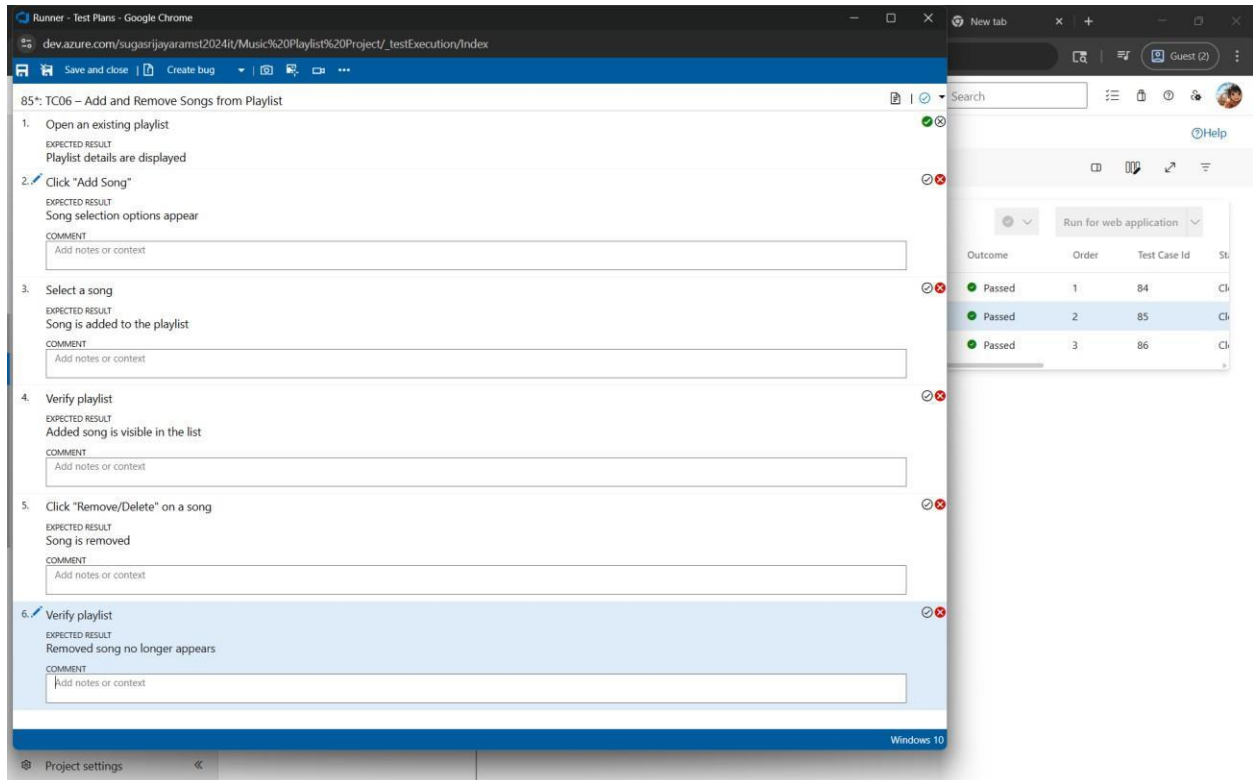
85: TC06 - Add and Remove Songs from Playlist

- Open an existing playlist
EXPECTED RESULT: Playlist details are displayed
ACTUAL RESULT: [Pass]
- Click "Add Song"
EXPECTED RESULT: Song selection options appear
ACTUAL RESULT: [Pass]
- Select a song
EXPECTED RESULT: Song is added to the playlist
ACTUAL RESULT: [Pass]
- Verify playlist
EXPECTED RESULT: Added song is visible in the list
ACTUAL RESULT: [Pass]
- Click "Remove/Delete" on a song
EXPECTED RESULT: Song is removed
ACTUAL RESULT: [Pass]
- Verify playlist
EXPECTED RESULT: Removed song no longer appears
ACTUAL RESULT: [Pass]

Run options:

- Run for web application
- Run for desktop application
- Run with options

5. Creating the bug



Runner - Test Plans - Google Chrome

dev.azure.com/sugasrijayaramst2024it/Music%20Playlist%20Project/_testExecution/Index

Save and close | Create bug

85: TC06 - Add and Remove Songs from Playlist

1. Open an existing playlist
EXPECTED RESULT
Playlist details are displayed
2. Click "Add Song"
EXPECTED RESULT
Song selection options appear
COMMENT
Add notes or context
3. Select a song
EXPECTED RESULT
Song is added to the playlist
COMMENT
Add notes or context
4. Verify playlist
EXPECTED RESULT
Added song is visible in the list
COMMENT
Add notes or context
5. Click "Remove/Delete" on a song
EXPECTED RESULT
Song is removed
COMMENT
Add notes or context
6. Verify playlist
EXPECTED RESULT
Removed song no longer appears
COMMENT
Add notes or context

Windows 10

Project settings

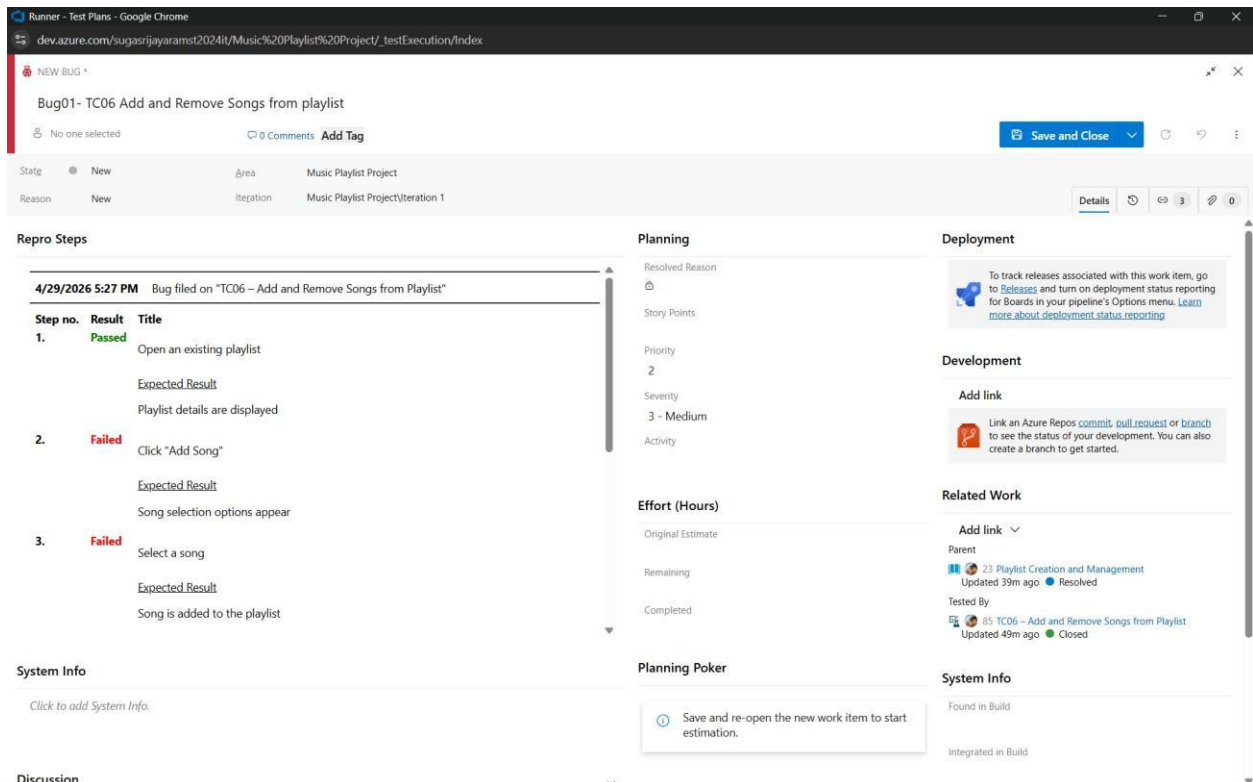
New tab

Guest (2)

Search

Run for web application

Outcome	Order	Test Case Id	St
Passed	1	84	Cl
Passed	2	85	Cl
Passed	3	86	Cl



Runner - Test Plans - Google Chrome

dev.azure.com/sugasrijayaramst2024it/Music%20Playlist%20Project/_testExecution/Index

NEW BUG

Bug01- TC06 Add and Remove Songs from playlist

No one selected | 0 Comments | Add Tag

Save and Close

State: New | Area: Music Playlist Project | Reason: New | Iteration: Music Playlist Project/Iteration 1

Details | 0 | 0 | 0

Repro Steps

4/29/2026 5:27 PM Bug filed on "TC06 - Add and Remove Songs from Playlist"

Step no.	Result	Title
1.	Passed	Open an existing playlist Expected Result Playlist details are displayed
2.	Failed	Click "Add Song" Expected Result Song selection options appear
3.	Failed	Select a song Expected Result Song is added to the playlist

System Info

Click to add System Info.

Discussion

Planning

Resolved Reason

Story Points

Priority

2

Severity

3 - Medium

Activity

Effort (Hours)

Original Estimate

Remaining

Completed

Planning Poker

Save and re-open the new work item to start estimation.

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also create a branch to get started.

Related Work

Add link

Parent

23 Playlist Creation and Management
Updated 39m ago | Resolved

Tested By

85 TC06 - Add and Remove Songs from Playlist
Updated 49m ago | Closed

System Info

Found in Build

Integrated in Build

6. Test case results

The screenshot shows the Azure DevOps interface for a project named 'Music Playlist Project'. The 'Test Plans' section is active, displaying a test plan titled '20: User Registration and Authentication'. The test plan status is '85% run, 100% passed'. A modal window titled 'Test point ID 2 execution history' is open, showing a table of test point execution results.

Outcome	Test run	Run by	Time completed
Passed	21	SUGASRIJAYARAM S...	19m ago
Passed	9	SUGASRIJAYARAM S...	Yesterday
Failed	8	SUGASRIJAYARAM S...	Yesterday
Passed	5	sriram S	Yesterday

Currently showing a test point execution history. View all history for a full test case execution history. [View all history](#)

7. Bug report

- Assigning bug to the developer and changing state to Active

The screenshot shows the Azure DevOps interface for a bug report titled 'Bug01 - TC06 Add and Remove Songs from Playlist'. The bug is currently in the 'Active' state. The 'Reason' dropdown is open, showing options: 'New', 'Active', 'Resolved', and 'Closed'. The 'Step no.' column shows three steps, with the first two marked as 'Failed'.

Step no.	Reason	Expected Result
1.	Failed	Playlist details are displayed
2.	Failed	Click "Add Song"
3.	Failed	Song selection options appear

System Info: Found in Build, Integrated in Build

BUG 87

87 Bug01 - TC06 Add and Remove Songs from Playlist

SUGASRIJAYARAM S T 0 Comments Add Tag

Save Follow

Updated by SUGASRIJAYARAM S T: Just now

Details 0

Stat: Active Area: Music Playlist Project Reason: Regression Iteration: Music Playlist Project/Iteration 1

4/29/2026 5:30 PM Bug filed on "TC06 - Add and Remove Songs from Playlist"

Step no.	Result	Title
1.	Passed	Open an existing playlist <u>Expected Result</u> Playlist details are displayed
2.	Failed	Click "Add Song" <u>Expected Result</u> Song selection options appear
3.	Failed	Select a song <u>Expected Result</u> Song is added to the playlist

System Info

Discussion

Story Points

Priority

2

Severity

3 - Medium

Activity

Effort (Hours)

Original Estimate

Remaining

Completed

Planning Poker

Voted: 0

0 1 2 3 5 8 13

Write your comment here

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

Parent

23 Playlist Creation and Management
Updated 58m ago Resolved

Tested By

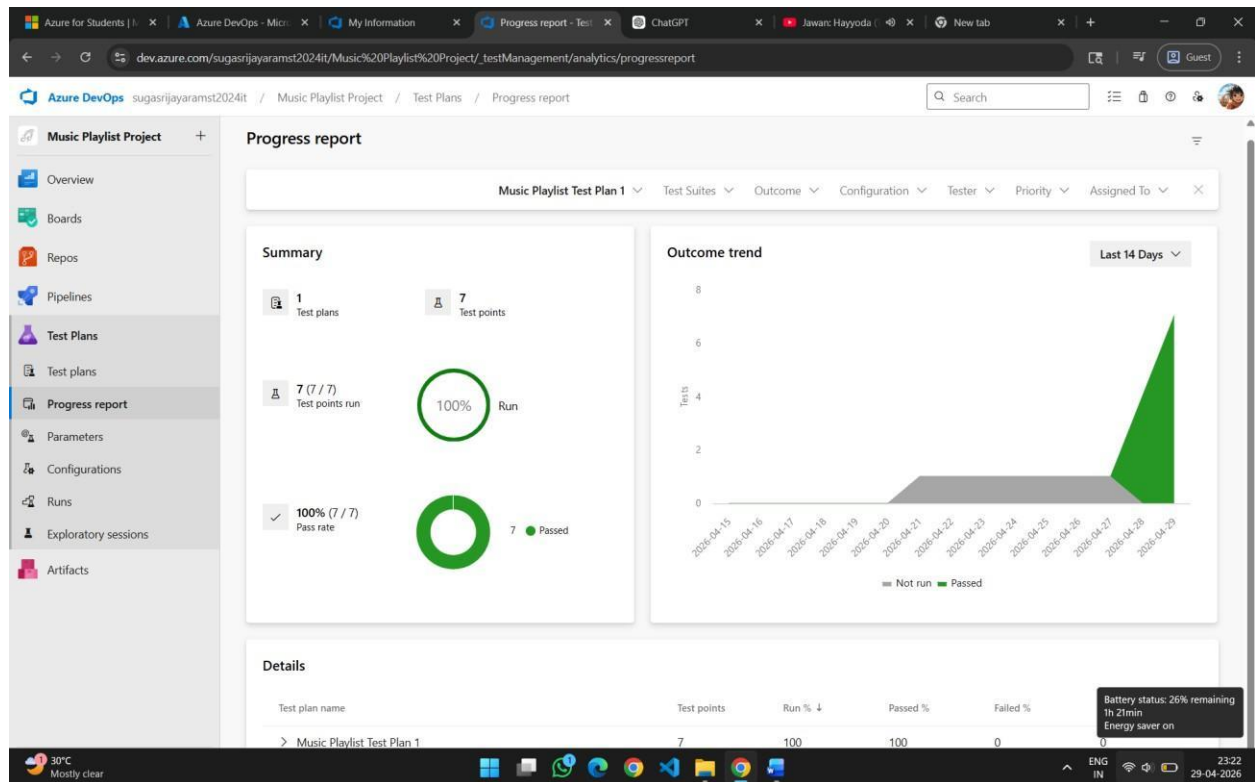
85 TC06 - Add and Remove Songs from Playlist
Updated 1h ago Closed

System Info

Found in Build

Integrated in Build

8.Progress report



Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path.

EXP NO: 9	LOAD TESTING AND PIPELINES
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Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests, and deployment.

Load Testing

Azure Load Testing:

Azure Load Testing allows you to simulate high traffic and stress tests for your web applications and APIs to understand how they perform under load. It helps identify performance bottlenecks, scalability issues, and optimize resource usage before deployment.

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

1. Sign in to Azure Portal
Go to <https://portal.azure.com> and log in.
2. Create the Resource
 - o Go to *Create a resource* → Search for “Azure Load Testing”.
 - o Select Azure Load Testing and click Create.
3. Fill in the Configuration Details
 - o *Subscription*: Choose your Azure subscription.
 - o *Resource Group*: Create new or select an existing one.
 - o *Name*: Provide a unique name (no special characters).
 - o *Location*: Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
5. Click Review + Create, then Create.
6. Once deployment is complete, click Go to resource.

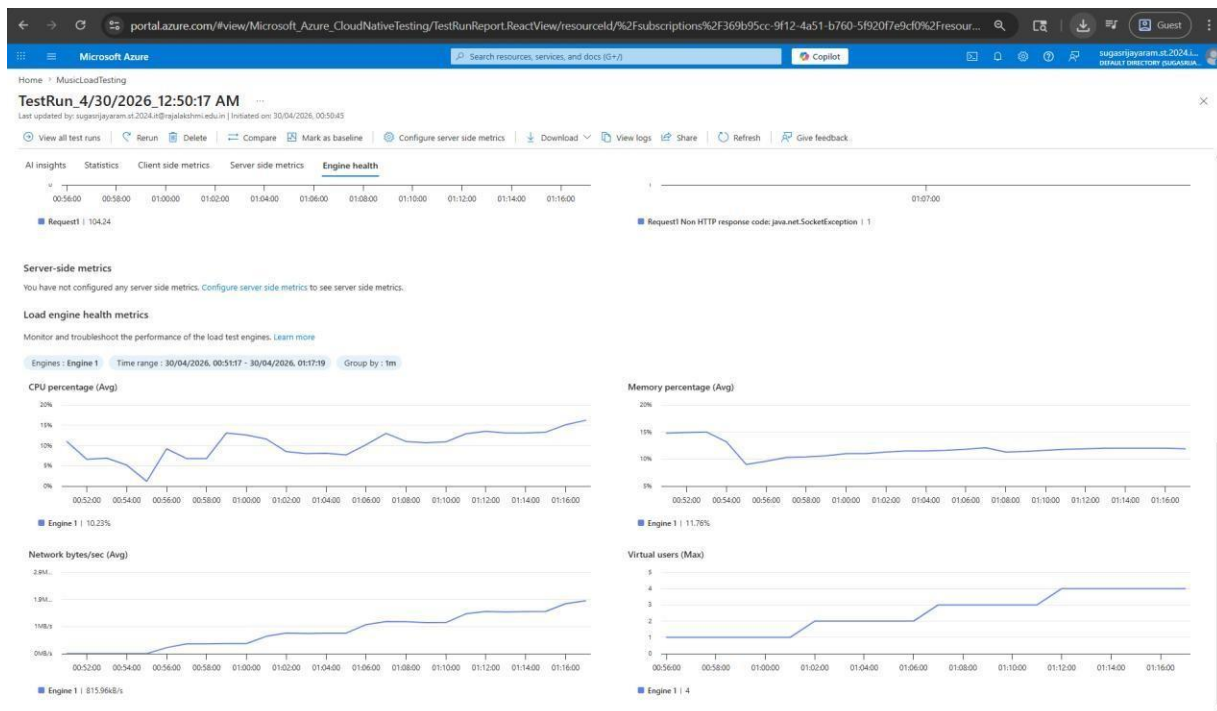
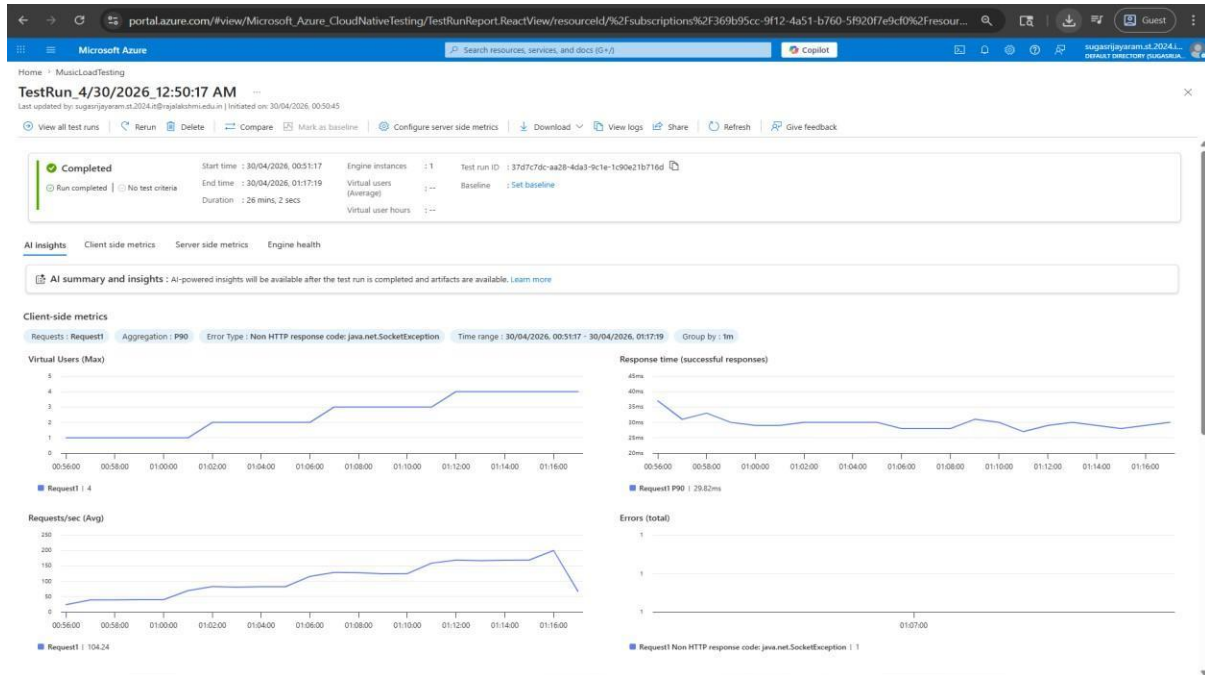
Steps to Create and Run a Load Test:

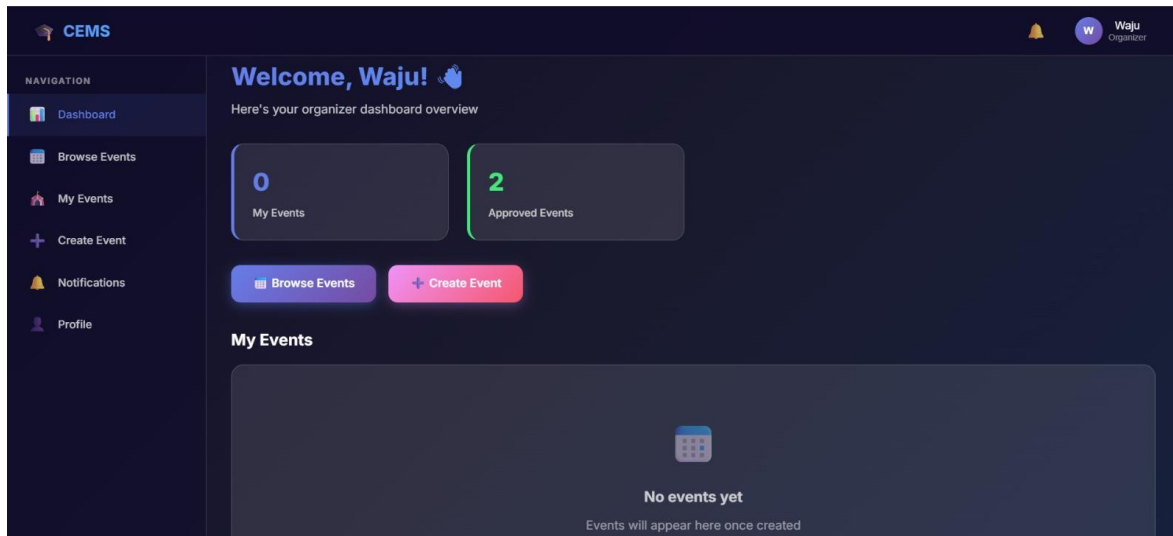
Once your resource is ready:

1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
2. Basics Tab
 - o *Test Name*: Provide a unique name.
 - o *Description*: (Optional) Add test purpose.
 - o *Run After Creation*: Keep checked.
3. Load Settings
 - o *Test URL*: Enter the target endpoint (e.g., <https://yourapi.com/products>).

4. Click Review + Create → Create to start the test.

Load Testing





Pipelines

Description:

This experiment demonstrates how to connect a GitHub-hosted Flask-based music recommendation project with Azure DevOps. The pipeline will automatically install dependencies, run basic tests, and publish artifacts. This ensures that every commit triggers checks for reliability and smooth deployment.

Steps:

1. Connect GitHub to Azure DevOps:
 - In Azure DevOps, create a new project.
 - Create a pipeline and select GitHub as the source.
 - Authorize access to your GitHub repository, ensuring that Azure DevOps can pull the repository for your pipeline.
2. Create azure-pipelines.yml in Your Repo Root:
 - In your GitHub repository, create a new file called azure-pipelines.yml in the root directory.
 - Add the following basic pipeline configuration for Python and Flask:

yaml Code

trigger:

- main

pool:

- vmImage: 'ubuntu-latest'

variables:

- NODE_VERSION: '20.19.x'

- NPM_CACHE: '\${Pipeline.Workspace}/.npm'

steps:

♦ Install Node.js (UPDATED TASK)

- task: UseNode@1

- inputs:

- version: \$(NODE_VERSION)

- displayName: 'Install Node.js'

♦ Cache npm (FIXED PATH)

- task: Cache@2

- inputs:

- key: 'npm | "\$(Agent.OS)" | **/package-lock.json'

- path: \$(NPM_CACHE)

- displayName: 'Cache npm'

♦ Install Frontend Dependencies

- script: |

- npm config set cache \$(NPM_CACHE) --global

- npm ci

- workingDirectory: frontend

- displayName: 'Install Frontend Dependencies'

♦ Build Frontend (Vite)

- script: npm run build

- workingDirectory: frontend

- displayName: 'Build Frontend'

♦ Frontend Tests (optional)

```
- script: npm test --if-present
  workingDirectory: frontend
  displayName: 'Run Frontend Tests'
```

♦ Install Backend Dependencies

```
- script: |
  npm config set cache $(NPM_CACHE) --global
  npm ci
  workingDirectory: backend
  displayName: 'Install Backend Dependencies'
```

♦ Backend Tests

```
- script: npm test --if-present
  workingDirectory: backend
  displayName: 'Run Backend Tests'
```

♦ Verify Frontend Build (fail early)

```
- script: |
  if [ ! -d "frontend/dist" ]; then
    echo "Build failed: dist folder not found"
    exit 1
  fi
  displayName: 'Verify Frontend Build'
```

♦ Archive Frontend

```
- task: ArchiveFiles@2
  inputs:
    rootFolderOrFile: 'frontend/dist'
    includeRootFolder: false
    archiveType: 'zip'
    archiveFile: '$(Build.ArtifactStagingDirectory)/frontend.zip'
  displayName: 'Archive Frontend'
```

♦ Archive Backend

```
- task: ArchiveFiles@2
  inputs:
    rootFolderOrFile: 'backend'
    includeRootFolder: false
    archiveType: 'zip'
    archiveFile: '$(Build.ArtifactStagingDirectory)/backend.zip'
  displayName: 'Archive Backend'
```

♦ Publish Artifacts

- task: PublishBuildArtifacts@1

inputs:

PathToPublish: '\$(Build.ArtifactStagingDirectory)'

ArtifactName: 'drop'

displayName: 'Publish Artifacts'

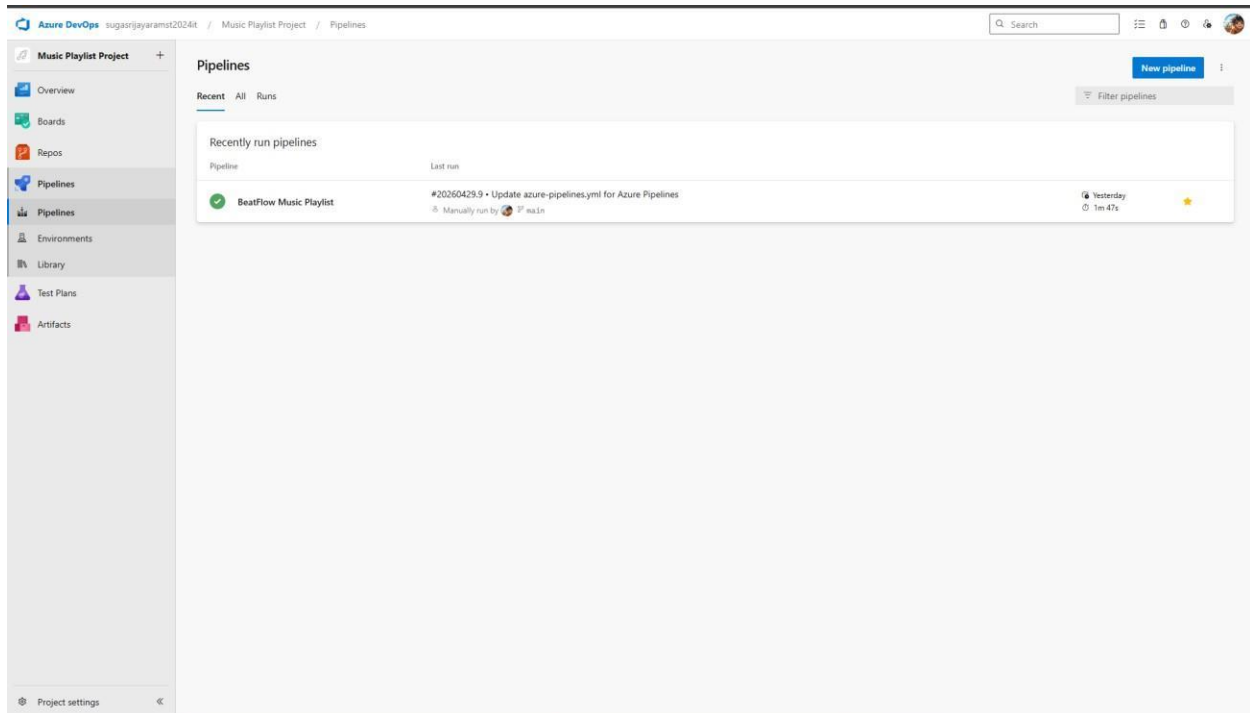
3. Run and Monitor Pipeline:

- Commit changes to the main branch of your repository to trigger the pipeline in Azure DevOps.
- Monitor the logs in the Azure DevOps portal to view logs, errors, or success messages and ensure everything runs smoothly.

Pipeline

The screenshot shows the Azure DevOps web interface for reviewing a pipeline YAML file. The breadcrumb navigation at the top indicates the path: **Azure DevOps** > *sugarsjaramst2024it* > *Music Playlist Project* > **Pipelines**. The left sidebar contains navigation links for Overview, Boards, Repos, Pipelines (selected), Environments, Library, Test Plans, and Artifacts. The main content area is titled "Review your pipeline YAML" and displays the YAML configuration for a pipeline named "BeatFlow Music Playlist / azure-pipelines-1.yml". The pipeline is a "Starter pipeline" with a trigger on the "main" branch. It uses an Ubuntu-latest pool and defines variables for Node.js version and npm cache. The steps include installing Node.js, caching npm, installing frontend dependencies, and building the frontend with Vite.

```
1 # Starter pipeline
2 # Start with a minimal pipeline that you can customize to build and deploy your code.
3 # Add steps that build, run tests, deploy, and more:
4 # https://aka.ms/yaml
5
6 trigger:
7   - main
8
9 pool:
10  - vmImage: 'ubuntu-latest'
11
12 variables:
13   - name: NODE_VERSION
14     value: '20.19.x'
15   - name: NPM_CACHE
16     value: '$(Pipeline.Workspace)/.npm'
17
18 steps:
19   - task: Install Node.js (UPDATED TASK)
20     inputs:
21       version: '$(NODE_VERSION)'
22       displayName: 'Install Node.js'
23
24   - task: Cache npm (FIXED PATH)
25     inputs:
26       key: 'npm | "$(Agent.OS)" | **/package-lock.json'
27       path: '$(NPM_CACHE)'
28       displayName: 'Cache npm'
29
30   - task: Install Frontend Dependencies
31     script: |
32       npm config set cache $(NPM_CACHE) --global
33       npm ci
34       workingDirectory: frontend
35       displayName: 'Install Frontend Dependencies'
36
37   - task: Build Frontend (Vite)
38     script: npm run build
39     workingDirectory: frontend
40     displayName: 'Build Frontend'
```



Result:

Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in azure devops.

EXP NO: 10

GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

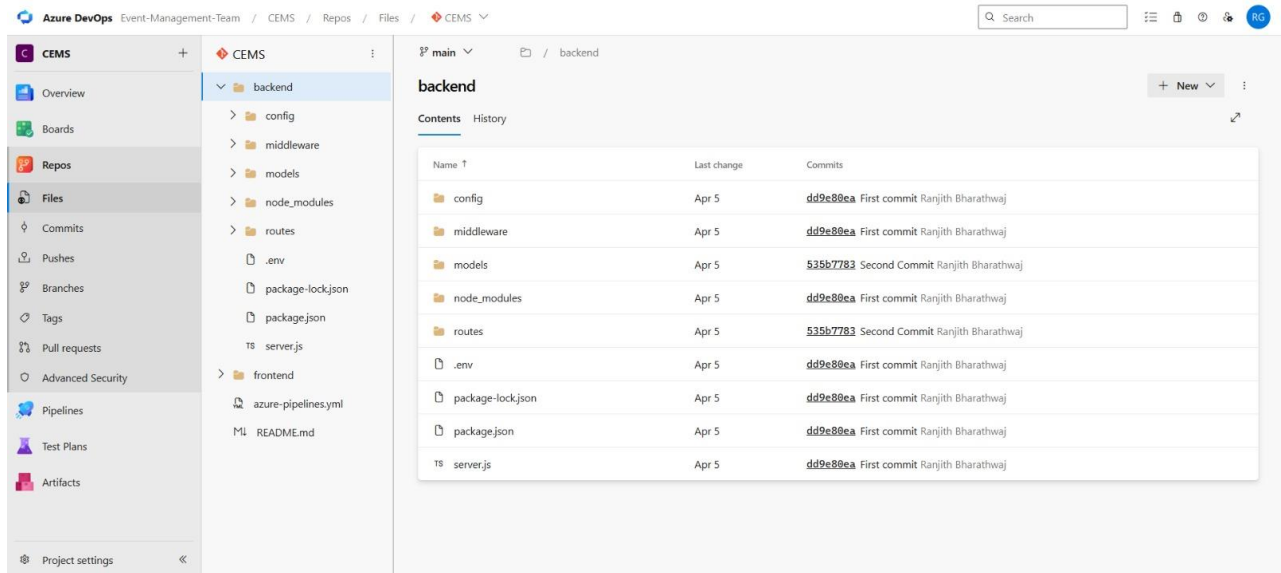
To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the College Event Management.

GitHub Project Structure

The screenshot displays the GitHub web interface for a repository named 'CEMS'. The left sidebar contains navigation links: Overview, Boards, Repos, Files, Commits, Pushes, Branches, Tags, Pull requests, Advanced Security, Pipelines, Test Plans, and Artifacts. The 'Files' section is active, showing a tree view with folders 'backend' and 'frontend', and files 'azure-pipelines.yml' and 'README.md'. The main content area shows the 'Files' tab with a table of repository contents:

Name	Last change	Commits
backend	Apr 5	535b7783 Second Commit Ranjith Bharathwaj
frontend	Apr 5	dd9e88ea First commit Ranjith Bharathwaj
azure-pipelines.yml	Apr 5	dd9e88ea First commit Ranjith Bharathwaj
README.md	Apr 5	dd9e88ea First commit Ranjith Bharathwaj

Below the table, the repository description reads: 'College Event Management System (CEMS) A full-stack web application for managing college events built with React, Node.js/Express, and MongoDB.' It also lists the technologies used: React 18, Node.js 16, MongoDB, and Azure DevOps CI/CD. A 'Features' section follows, detailing: User Management (Registration, login, profile management with JWT auth), Role-Based Access (Student, Organizer, Admin roles with different permissions), Event Management (Create, edit, delete events with admin approval workflow), Event Discovery (Browse, search, and filter events by category), Registration System (Register for events with duplicate prevention), Notifications (Real-time notifications for registrations and event updates), and Admin Dashboard (Reports, user management, event approvals).



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.